DIGITAL SYSTEMS AND COMPUTER ORGANIZATION — CES 2104

Chapterwise Previous Year Questions

Source papers. 48 papers — CSS 2104 Digital Systems and Computer Organization (Nov 2025, Dec 2025) · ICT 2123 / IT 2123 Digital Systems and Computer Organization (2023–2025) · ECM 2122 Digital System Design and Computer Organization (2024–2025) · CSE 2121 / CSE 2151 / ECE 2152 Computer Organization and Architecture (2022–2026) · CSE 2123 / CSE 2153 / ECE 2124 / ECE 2153 / DS 2123 Digital System Design (2022–2025) · ICT 2154 Digital Systems (2022–2023) · ICT 2256 / IT 2256 Computer Organization and Microprocessor Systems (2023–2024)

About this document. CES 2104 is a **merger of two older subjects** — "Digital System Design" (which maps onto Modules 1–4) and "Computer Organization and Architecture" (Modules 5–9). That is why this subject has by far the most source papers, and why any single old paper covers only about half your syllabus. CSS 2104 is the direct predecessor and is the only code that covers all nine modules in one paper. ECE-variant papers are tagged; they cover the same digital-design ground but substitute FPGA and VLSI content for the computer-organization half.

A note on figures. Questions marked [figure/table not in text] had their state diagram, circuit, K-map, register transfer description or truth table embedded as an image in the original PDF. The wording is complete; only the visual is missing. Open the source paper for those.

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Module 1 — Overview of K-Map

Logic gates and truth tables · K-map simplification (SOP and POS) · incompletely specified functions · multilevel NAND and NOR circuits · introduction to Verilog HDL

Q1. **[5M]** *IT 2123 · Nov 2025* Simplify a 4-variable Boolean function using a K-map and implement using NAND gates only. *(K-map + NAND/NOR only asked 7×)*

Q2. **[3M]** *IT 2123 · Nov 2025* Design a circuit for F(A,B,C,D) = Σ(0,1,2,5,8,9,10) using a 4-variable K-map. *(K-map + NAND/NOR only asked 7×)*

Q3. **[3M]** *CSS 2104 · Nov 2025* Simplify the given function and implement it using only NOR gates. [figure/table not in text] *(K-map + NAND/NOR only asked 7×)*

Q4. **[2M]** *CSE 2123 · Dec 2025* Simplify the function F(a,b,c,d) = ΠM(1,3,5,8,9,11,15) + D(2,13) using a K-map to obtain the minimum-cost SOP expression and draw the circuit. *(K-map + NAND/NOR only asked 7×)*

Q5. **[3M]** *CSE 2153 · Dec 2022* Simplify the function F(a,b,c,d) = ΠM(1,3,5,8,9,11,15) + D(2,13) using a K-map to obtain the minimum-cost SOP expression and draw its NAND-only circuit. *(K-map + NAND/NOR only asked 7×)*

Q6. **[2M]** *CSE 2123 · Dec 2024* Simplify the given function using a 4-variable K-map, then implement the simplified expression using only NAND gates. [figure/table not in text] *(K-map + NAND/NOR only asked 7×)*

Q7. **[4M]** *ECE 2153 · Dec 2022* What are universal gates? Simplify the following Boolean expression using a Karnaugh map and draw the logic circuit using only NAND gates: F(A,B,C,D) = Σm(6,7,8,9,10,12) + d(2,5,13). — *ECE-variant (K-map + NAND/NOR only asked 7×)*

Q8. **[3M]** *CSE 2153 · Jan/Feb 2023* Simplify the following function as a sum of products using the K-map technique: f(A,B,C,D) = Σm(0,1,4,8,9,10).

Q9. **[2M]** *ICT 2123 · Dec 2024* For a function defined by minterms 1, 3, 7, 11 and 15 with don't-cares at 0, 2, 4 and 5, fill the K-map and minimise the function. *(don't-care conditions asked 3×)*

Q10. **[2M]** *CSE 2123 · Nov 2025* Analyze the Boolean function f(x1,x2,x3,x4) = x1'x3'x4' + x3x4 + x1'x2'x4 + x1x2x3'x4 with the don't-care set D = Σ(9, 12, 14) to find the minimum-cost SOP and POS expressions. *(don't-care conditions asked 3×)*

Q11. **[2M]** *ICT 2123 · Nov/Dec 2024* Simplify the Boolean expression into POS and implement using NOR gates: F(A,B,C,D) = Σ(1,5,6,7,8,9,10,14). *(POS implementation asked 3×)*

Q12. **[3M]** *CSE 2153 · Jan 2024* Find the POS implementation for the given function and compare the cost with the SOP form. [figure/table not in text] *(POS implementation asked 3×)*

Q13. **[3M]** *ECE 2153 · Dec 2022* Reduce the function F(A,B,C) = ΠM(0,1,2,3,4,7) using a K-map and implement the same using only NOR gates. — *ECE-variant (POS implementation asked 3×)*

Q14. **[3M]** *ECM 2122 · 2024 Regular* Reduce the following 5-variable Boolean function in SOP form using the five-variable K-map: F(A,B,C,D,E) = Σm(0,1,4,5,6,13,14,15,22,24,25,28,29,30,31). *(5-variable K-map asked 3×)*

Q15. **[2M]** *ICT 2154 · Jan/Feb 2023* Solve the SOP function minimisation of a 5-variable K-map: F(A,B,C,D,E) = Σ(0,2,4,6,9,11,13,15,17,21,25,27,29,31). *(5-variable K-map asked 3×)*

Q16. **[2M]** *ICT 2154 · Dec 2022* Solve the POS function minimisation of a 5-variable K-map: ΠM(0,2,4,7,8,10,12,16,18,20,23,24,25,26,27,28). *(5-variable K-map asked 3×)*

Q17. **[3M]** *CSE 2153 · Nov/Dec 2023* Make use of functional decomposition to find the minimum-cost circuit for the function f(x1,x2,x3,x4) = Σm(4,7,9,10,12,13,14,15). Assume the input variables are available in uncomplemented form only. *(functional decomposition asked 2×)*

Q18. **[3M]** *CSE 2123 · Nov/Dec 2023* Make use of functional decomposition to find the minimum-cost circuit for f(x1,x2,x3,x4) = Σm(4,7,9,10,12,13,14,15), with input variables available in uncomplemented form only. *(functional decomposition asked 2×)*

Q19. **[5M]** *IT 2123 · Dec 2025* Explain how an XOR gate can be realised using only NAND gates with a neat diagram.

Q20. **[4M]** *ECE 2124 · (undated)* Design a combinational logic circuit with 4 inputs (A,B,C,D) and one output (Y) that produces a high output if any three inputs are high. Implement using the minimum number of NAND gates. — *ECE-variant*

Q21. **[2M]** *CSE 2123 · Nov/Dec 2024* Draw a logic diagram using 2-input NOR gates only to implement the given function. [figure/table not in text]

Q22. **[3M]** *ECE 2124 · Nov/Dec 2024* A staircase light is controlled by two switches, one at the top and one at the bottom of the stairs. The light is ON when and only when one switch is ON and the other is OFF. Make a truth table for this system and implement it. — *ECE-variant*

Introduction to Verilog HDL

Q23. **[3M]** *IT 2123 · Dec 2025* Write a Verilog code snippet for a 2-input OR gate using dataflow modeling.

Q24. **[2M]** *ECM 2122 · Dec 2025* With an example, explain the role of instances in Verilog programming.

Q25. **[2M]** *ECM 2122 · Nov 2025* Differentiate between a hardware description language (such as Verilog) and C language.

Q26. **[2M]** *CSE 2123 · Nov 2025* Distinguish between blocking and non-blocking assignments in Verilog with an example. *(blocking vs non-blocking asked 2×)*

Q27. **[3M]** *CSE 2153 · Dec 2022* Distinguish between blocking and non-blocking assignments in Verilog. Draw the circuits for the given codes. [figure/table not in text] *(blocking vs non-blocking asked 2×)*

Q28. **[3M]** *ECE 2124 · Nov/Dec 2023* Write the Verilog code for a 3-input AND gate as a user-defined primitive (UDP). — *ECE-variant (UDP asked 6×)*

Q29. **[3M]** *ECE 2124 · Jan 2024* Write the Verilog code for a 3-input XOR gate as a user-defined primitive (UDP). — *ECE-variant (UDP asked 6×)*

Q30. **[3M]** *ECE 2124 · Dec 2024* Write a Verilog code for a 3-input XOR gate using a user-defined primitive (UDP). — *ECE-variant (UDP asked 6×)*

Q31. **[3M]** *ECE 2124 · Nov/Dec 2024* Write the Verilog code for a 3-input NOR gate using a user-defined primitive. — *ECE-variant (UDP asked 6×)*

Q32. **[3M]** *ECE 2124 · (undated)* Write a Verilog code to implement a 3-bit even parity generator using a user-defined primitive (UDP). — *ECE-variant (UDP asked 6×)*

Q33. **[3M]** *ECE 2153 · Dec 2022* Write the Verilog code for a 3-input NAND gate using a user-defined primitive. — *ECE-variant (UDP asked 6×)*

Q34. **[4M]** *ECE 2124 · Nov 2025* Write a Verilog user-defined primitive (UDP) for the carry-out output of a full adder circuit. — *ECE-variant*

Module 2 — Arithmetic Circuits

Addition of unsigned numbers · half adder · full adder · ripple carry adder · signed numbers · adder/subtractor · BCD adder · array multiplier · design of arithmetic circuits using Verilog

Half and full adders

Q1. **[5M]** *IT 2123 · Dec 2025* Design a Half Adder and Full Adder using logic gates and verify their operation. *(full adder design asked 6×)*

Q2. **[5M]** *ECM 2122 · 2024 Regular* With a neat diagram and truth table, implement a full adder using two half adders. Also implement the same with a 3×8 decoder. *(full adder design asked 6×)*

Q3. **[3M]** *ECE 2124 · Nov/Dec 2023* Implement a full adder using two half-adder circuits and one OR gate. — *ECE-variant (full adder design asked 6×)*

Q4. **[3M]** *IT 2123 · Nov/Dec 2023* Design a full adder combinational circuit using a decoder. *(full adder design asked 6×)*

Q5. **[3M]** *IT 2123 · Jan 2024* Design a Full Adder using active-low output 2-to-4 decoders with enable and external gates. *(full adder design asked 6×)*

Q6. **[2M]** *ICT 2154 · Jan/Feb 2023* Design and implement a full adder circuit using a 3:8 decoder. *(full adder design asked 6×)*

Q7. **[3M]** *ECE 2153 · Jan/Feb 2023* Implement the full adder using an active-low decoder. — *ECE-variant*

Q8. **[3M]** *ECE 2124 · (undated)* Design a full adder circuit using two 2:1 multiplexers and additional gates by considering the MSB of inputs as the selection line. — *ECE-variant*

Q9. **[4M]** *ECE 2124 · Nov/Dec 2024* Write structural Verilog code for a full adder with two half-adder modules and logical gates. — *ECE-variant*

Q10. **[4M]** *ECE 2124 · Jan 2024* Write the structural Verilog code for a 4-bit parallel adder. — *ECE-variant*

Q11. **[3M]** *ECM 2122 · 2024 Regular* Write a Verilog code to implement a four-bit parallel binary adder using a full adder.

Q12. **[3M]** *ECE 2124 · Jan 2024* Design a full subtractor using a decoder and logic gates. Explain its working principle. — *ECE-variant (full subtractor asked 4×)*

Q13. **[3M]** *ICT 2154 · Jan/Feb 2023* Design a full subtractor using 4:1 MUXs only. *(full subtractor asked 4×)*

Q14. **[3M]** *ECE 2124 · (undated)* Write dataflow modeling in Verilog to implement a full subtractor. — *ECE-variant (full subtractor asked 4×)*

Q15. **[3M]** *ECE 2124 · Nov 2025* Write a structural Verilog module for a full subtractor by hierarchically integrating half subtractors and the necessary logic gates. — *ECE-variant (full subtractor asked 4×)*

Q16. **[4M]** *ECM 2122 · Dec 2025* Design a combinational circuit using a programmable logic array (PLA) to implement a full subtractor.

Carry lookahead adders

Q17. **[5M]** *IT 2123 · Nov/Dec 2023* Design a 4-bit parallel adder with carry lookahead adder. *(carry lookahead adder asked 8×)*

Q18. **[5M]** *ICT 2154 · Jan/Feb 2023* Design a 4-bit Carry Look Ahead binary adder using logic gates. *(carry lookahead adder asked 8×)*

Q19. **[5M]** *ICT 2123 · Nov/Dec 2024* Design a 4-bit Carry Lookahead Adder (CLA) to add A = 1101_2 and B = 1011_2. (a) Calculate the generate (G) and propagate (P) signals for each bit. (b) Determine the carry outputs for each bit. (c) Calculate the final sum. *(carry lookahead adder asked 8×)*

Q20. **[5M]** *CSE 2123 · Dec 2025* Apply the concept of the Carry-Lookahead Adder by deriving the general expressions for Carry Generate (G_i), Carry Propagate (P_i) and the output carry (C_{i+1}) for a single stage. Apply these to write the specific logical expressions for C_1 and C_2 in terms of the initial carry C_0. *(carry lookahead adder asked 8×)*

Q21. **[3M]** *CSE 2123 · Dec 2024* Design and develop a 4-bit carry-lookahead adder. Apply a suitable approach to optimise the design to reduce the delay. *(carry lookahead adder asked 8×)*

Q22. **[3M]** *CSE 2153 · Dec 2022* Design a 2-bit Carry Look-Ahead adder using logic gates. *(carry lookahead adder asked 8×)*

Q23. **[3M]** *CSE 2153 · Jan/Feb 2023* Design a 48-bit carry lookahead adder with ripple carry between 8-bit carry lookahead blocks. Determine the gate delay involved to generate the final carry. *(carry lookahead adder asked 8×)*

Q24. **[4M]** *CSE 2153 / CSE 2123 · Jan 2024* Design a hierarchical 40-bit carry look-ahead adder with ripple-carry between 8-bit blocks. Compute the total gate delay for the design, mentioning all intermediary gate delays. *(carry lookahead adder asked 8×)*

Q25. **[5M]** *ECE 2152 · Dec 2022* Explain the working of a carry lookahead adder (CLA) circuit. Using a 4-bit CLA as a building block, design a 16-bit CLA. Also calculate the worst-case add time of the 16-bit CLA. — *ECE-variant*

Q26. **[4M]** *ECE 2152 · Nov/Dec 2023* Explain the working of carry lookahead adder, carry select adder and carry save adder in detail. — *ECE-variant*

Adder/subtractor and signed arithmetic

Q27. **[4M]** *CSE 2153 · Nov/Dec 2023* Design a combinational unit that can act as adder and subtractor for two 4-bit numbers. Provide suitable explanation. *(adder/subtractor asked 3×)*

Q28. **[4M]** *ECE 2124 · Nov/Dec 2024* Design a 4-bit ripple carry adder/subtractor using a full adder and necessary gates. Explain the working principle with suitable examples. — *ECE-variant (adder/subtractor asked 3×)*

Q29. **[5M]** *CSE 2123 · Dec 2024* Design a simple processor with control circuitry that can perform the addition/subtraction of two 4-bit data. Provide suitable explanation. *(adder/subtractor asked 3×)*

Q30. **[5M]** *IT 2123 · Jan 2024* If X = 0, F = Y+1, otherwise F = Y−1, where X is a single-bit number and Y and F are 4-bit binary numbers. Design the logic circuit using (i) only a single 7483 IC (ii) the minimum number of half adders and one external gate only.

Q31. **[3M]** *ECE 2153 · Dec 2022* Perform a 3-bit subtraction of 101−011 using a 3-bit parallel binary adder/subtractor circuit. — *ECE-variant*

Q32. **[5M]** *ICT 2154 · Dec 2022* Design a combinational circuit to perform F = A−B when control input Y = 0 and F = 1+2A−B when Y = 1, using 4-bit parallel binary adders and minimum universal gates. Assume A, B and F are 4-bit binary numbers.

Q33. **[5M]** *ICT 2123 · Dec 2024* Design a combinational circuit using a 4-bit adder and external gates to add two 4-bit numbers A and B, where: if C < 3, B = 2C; if C is a multiple of 3, B = C+4; otherwise B = C, where C is a 3-bit binary number.

Q34. **[3M]** *ECM 2122 · 2024 Makeup* Design a digital circuit using a parallel adder and 2:1 MUX that works as specified. Assume A is 4-bit data. [figure/table not in text]

Q35. **[3M]** *CSS 2104 · (undated)* Represent the following numbers in 5-bit 2's complement form, perform the operation mentioned and state whether overflow occurs: (i) Add +9 and +12 (ii) Add −8 and −9 (iii) Subtract −5 from +10. *(2's complement overflow asked 6×)*

Q36. **[2M]** *CSE 2121 · Nov 2025* Convert each of the following pairs of decimal numbers into 6-bit 2's complement representation and perform the specified arithmetic operation. Determine whether overflow occurs and explain why: (i) Add 4 and 11 (ii) Subtract −9 from −2. *(2's complement overflow asked 6×)*

Q37. **[2M]** *CSE 2121 / CSE 2151 · Jan 2024* Convert the following pairs of decimal numbers to 6-bit 2's-complement numbers and perform the operation mentioned. State whether overflow occurs and why: (i) Add 4 and 11 (ii) Subtract −9 from −2. *(2's complement overflow asked 6×)*

Q38. **[3M]** *CSE 2151 · Jan/Feb 2023* Represent the following integers in 2's complement with the minimum number of bits required, perform the operation and state whether overflow occurs and why: (i) Add −8 with −7 (ii) Subtract +15 from −16. *(2's complement overflow asked 6×)*

Q39. **[3M]** *CSE 2151 · Dec 2022* Represent the following integers in 2's complement with minimum bits, perform the operation and state whether overflow occurs and why: (i) Add +8 with +8 (ii) Subtract −11 from −15. *(2's complement overflow asked 6×)*

Q40. **[2M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Add the following pairs of decimal numbers after representing them in 5-bit 2's complement format. Check and justify if overflow occurs: (i) 7 and 15 (ii) −5 and 9. *(2's complement overflow asked 6×)*

Q41. **[2M]** *CSE 2121 · Nov/Dec 2024* Represent the following integers in 2's complement with minimum bits, perform the operation and state whether overflow occurs and why: (a) Add −13 and 12 (b) Subtract −7 and 6.

Q42. **[2M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* State the arithmetic addition and subtraction rules of signed integers.

Q43. **[2M]** *IT 2123 · Nov 2025* Differentiate between signed and unsigned addition with a numerical example.

BCD adder

Q44. **[4M]** *IT 2123 · Nov 2025* Explain the design of a BCD adder and how it differs from a binary adder. *(BCD adder asked 3×)*

Q45. **[3M]** *ECE 2124 · (undated)* Design a BCD adder using two 4-bit adders and a correction detector circuit. Illustrate with suitable examples. — *ECE-variant (BCD adder asked 3×)*

Q46. **[3M]** *ECE 2124 · Nov 2025* Design a single-digit BCD adder using two 4-bit binary adders and realize the BCD correction circuit using NAND gates. Explain the operation with suitable examples. — *ECE-variant (BCD adder asked 3×)*

Multipliers

Q47. **[2M]** *IT 2123 · Nov/Dec 2023* Design a 4-bit by 3-bit binary multiplier and explain its operation. *(combinational multiplier asked 4×)*

Q48. **[3M]** *ICT 2154 · Jan/Feb 2023* Design a 3×2 bit combinational multiplier using basic gates. *(combinational multiplier asked 4×)*

Q49. **[3M]** *ICT 2154 · Dec 2022* Design a 2×2 bit combinational multiplier using basic gates. *(combinational multiplier asked 4×)*

Q50. **[3M]** *ICT 2123 · Dec 2024* Design a 2-bit × 2-bit binary multiplier using a 3-to-8 decoder with active low output and minimum external gates. *(combinational multiplier asked 4×)*

Q51. **[2M]** *CSE 2123 · Dec 2025* Illustrate a circuit using full adders to find the number of 1's in a three-bit unsigned number.

Module 3 — Combinational Circuit Building Blocks

Multiplexer · decoder · encoder · arithmetic comparison circuits · Verilog for combinational circuits

Multiplexers

Q1. **[3M]** *IT 2123 · Dec 2025* Implement a 4-to-1 Multiplexer using logic gates and derive the Boolean expression. *(MUX design asked 9×)*

Q2. **[3M]** *IT 2123 · Nov 2025* Implement a 16-to-1 Multiplexer using logic gates and derive the Boolean expression. *(MUX design asked 9×)*

Q3. **[3M]** *IT 2123 · Nov/Dec 2023* Design a multiplexer with a 16-to-1 configuration by using 2-to-1 multiplexers only. Illustrate the block diagram and truth table. *(MUX design asked 9×)*

Q4. **[3M]** *ICT 2154 · Dec 2022* Design a 16:1 MUX using 2:1 MUXs only. *(MUX design asked 9×)*

Q5. **[3M]** *ICT 2123 · Dec 2024* Show how to create an 8-to-1 multiplexer by combining two 4-to-1 multiplexers and an additional 2-to-1 multiplexer. *(MUX design asked 9×)*

Q6. **[3M]** *CSE 2123 · Dec 2025* Design a 4-to-1 multiplexer using only 2-to-1 multiplexers. Implement this design in Verilog HDL and simulate the circuit. *(MUX design asked 9×)*

Q7. **[3M]** *ECM 2122 · Dec 2025* Write Verilog dataflow code to implement an 8:1 multiplexer without using instances of 2:1 multiplexers. *(MUX design asked 9×)*

Q8. **[3M]** *ECM 2122 · Nov 2025* Write a Verilog structural code to implement a 4:1 multiplexer using instances of 2:1 multiplexers in dataflow. *(MUX design asked 9×)*

Q9. **[3M]** *ECM 2122 · 2024 Makeup* Write a Verilog code to implement an 8:1 mux using 2:1 mux. *(MUX design asked 9×)*

Q10. **[4M]** *ECE 2124 · Nov/Dec 2023* Write a structural Verilog code for an 8:1 multiplexer using a 2:1 multiplexer. — *ECE-variant (structural Verilog MUX asked 4×)*

Q11. **[3M]** *ECE 2124 · (undated)* Write a structural Verilog model for a 4-to-1 multiplexer using 2-to-1 multiplexer modules. — *ECE-variant (structural Verilog MUX asked 4×)*

Q12. **[3M]** *ECE 2153 · Jan/Feb 2023* Write a structural Verilog code for an 8-to-1 multiplexer using a 2-to-1 multiplexer. — *ECE-variant (structural Verilog MUX asked 4×)*

Q13. **[4M]** *ECE 2153 · Dec 2022* Write the structural Verilog code for an 8:1 multiplexer using a 2:1 multiplexer. — *ECE-variant (structural Verilog MUX asked 4×)*

Q14. **[3M]** *ECE 2124 · Nov/Dec 2023* Write dataflow Verilog code for a 4-to-1 multiplexer with delay. — *ECE-variant*

Q15. **[2M]** *ECM 2122 · Dec 2025* Write a behavioral Verilog code for a 4:1 MUX using an "always block" and "case statement".

Q16. **[3M]** *CSS 2104 · (undated)* Implement a three-input XOR gate using a 4-to-1 multiplexer with a neat diagram. *(MUX function realization asked 8×)*

Q17. **[3M]** *CSE 2123 · Nov 2025* Implement a three-input XOR gate using a 4-to-1 multiplexer with a neat diagram. *(MUX function realization asked 8×)*

Q18. **[3M]** *ICT 2123 · Nov/Dec 2024* Realize f1(x,y,z,w) = Σ(0,2,4,5,8,9,12,13) using an 8:1 multiplexer and basic gates. *(MUX function realization asked 8×)*

Q19. **[4M]** *ECE 2124 · Jan 2024* Implement the logic function using an 8×1 MUX: F(A,B,C,D) = Σm(1,3,4,11,12,13,14,15). — *ECE-variant (MUX function realization asked 8×)*

Q20. **[4M]** *ECE 2153 · Jan/Feb 2023* Implement the following logic function using (i) an 8:1 multiplexer (ii) a 2:1 multiplexer: F(A,B,C,D) = Σm(1,3,4,11,12,13,14,15). — *ECE-variant (MUX function realization asked 8×)*

Q21. **[3M]** *ECE 2124 · Dec 2024* Design the specified Boolean function using an 8:1 multiplexer, where D is the input and A, B, C are the selection lines: F(A,B,C,D) = AB' + BD + B'CD'. — *ECE-variant (MUX function realization asked 8×)*

Q22. **[5M]** *CSE 2123 · Nov 2025* Design the least-cost implementation with a 2:1 multiplexer and other necessary gates for the following function using Shannon's expansion: F(a,b,c,d) = abc + a'cd' + bcd. *(Shannon's expansion asked 2×)*

Q23. **[3M]** *CSE 2153 · Dec 2022* Design the least-cost implementation with a 2:1 multiplexer and other necessary gates for the following function using Shannon's expansion: F(a,b,c,d) = abc + a'cd' + bcd. *(Shannon's expansion asked 2×)*

Q24. **[2M]** *CSE 2123 · Nov/Dec 2024* Using a 2-to-1 multiplexer, design a combinational circuit with three inputs and one output in which the output will be 1 only when the input is even.

Q25. **[3M]** *ECE 2124 · Nov/Dec 2024* Implement a 2-bit equality detector using a 4:1 multiplexer and additional gates. — *ECE-variant*

Q26. **[5M]** *CSE 2123 · Nov 2025* Since a computer has many registers, paths must be provided to transfer information from one register to another. Design a 4-bit common bus system for 4 registers of 4 bits each using multiplexers. *(common bus system asked 4×)*

Q27. **[5M]** *CSE 2123 · Dec 2024* Design a 4-bit common bus system for 4 registers of 4 bits each using multiplexers. *(common bus system asked 4×)*

Q28. **[5M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Design a 4-bit common bus system for 4 registers of 4 bits each using multiplexers. *(common bus system asked 4×)*

Q29. **[5M]** *CSE 2123 · Nov 2025* Analyse the use of multiplexers to implement a bus. Describe the primary components of this bus structure and apply this structure to explain how data selection is performed. *(common bus system asked 4×)*

Decoders and demultiplexers

Q30. **[3M]** *ICT 2123 · Nov/Dec 2024* Design a 2-bit binary adder using 3-to-8 decoders with active-high outputs and external gates. *(decoder-based design asked 6×)*

Q31. **[2M]** *ICT 2154 · Dec 2022* Design a 4×16 decoder using only 3×8 decoders. *(decoder cascading asked 6×)*

Q32. **[3M]** *ECE 2124 · Nov/Dec 2023* Implement f(A,B,C,D) = Σ(0,3,5,9) using an optimised number of 2-to-4 decoders and additional gates. — *ECE-variant (decoder-based design asked 6×)*

Q33. **[3M]** *ECE 2124 · (undated)* Design a 4:16 decoder using 3:8 decoders with active-low enable input. — *ECE-variant (decoder cascading asked 6×)*

Q34. **[3M]** *ECE 2124 · Dec 2024* Implement a 4-to-16 decoder using 2-to-4 decoders using an optimised number of decoders. — *ECE-variant (decoder cascading asked 6×)*

Q35. **[3M]** *ECE 2124 · Nov/Dec 2024* Implement a 4-to-16 decoder using 3-to-8 decoders with active-low enable input signal. — *ECE-variant (decoder cascading asked 6×)*

Q36. **[3M]** *ECE 2124 · Nov 2025* Design a 4:16 line decoder with active-high enable and active-high outputs by using 2:4 line decoders and the necessary logic gates. — *ECE-variant (decoder cascading asked 6×)*

Q37. **[4M]** *ECE 2153 · Dec 2022* Implement a 1:16 demultiplexer using only 1:4 demultiplexers with a functional table. — *ECE-variant*

Q38. **[3M]** *ECE 2124 · Nov 2025* Implement the logic functions F1(A,B,C) = Σm(1,4,6), F2(A,B,C) = Σm(3,5) and F3(A,B,C) = Σm(0,2,7) using a 1-to-8 demultiplexer and necessary additional logic gates (DEMUX has active-low outputs). — *ECE-variant*

Q39. **[3M]** *CSE 2153 · Jan 2024* Implement the given function using a 3-to-8 decoder and an OR gate. [figure/table not in text]

Q40. **[3M]** *CSE 2153 · Nov/Dec 2023* A combinational circuit receives two single-bit inputs A and B and produces a single-bit output Y = F1|F2, where F1 and F2 are implemented using a decoder and OR gates. Design a minimal circuit for Y using only one decoder, one 2:1 multiplexer and OR gates.

Q41. **[3M]** *ECM 2122 · 2024 Makeup* Explain the operation of a 4-to-2 priority encoder using a neat diagram and truth table. *(priority encoder asked 5×)*

Q42. **[2M]** *CSE 2123 · Dec 2024* Develop a 4-to-2 priority encoder using Verilog HDL. Implement the circuit and simulate its output for different input combinations, ensuring the highest-priority input is encoded. *(priority encoder asked 5×)*

Q43. **[3M]** *ECE 2124 · Nov/Dec 2023* Write a sequential Verilog code for an 8:3 priority encoder using a case statement. — *ECE-variant (priority encoder asked 5×)*

Q44. **[4M]** *ECE 2124 · Nov 2025* Write behavioural Verilog coding for an 8:3 priority encoder using nested if statements. — *ECE-variant (priority encoder asked 5×)*

Q45. **[4M]** *ECE 2153 · Jan/Feb 2023* Write a sequential Verilog code for an 8-to-3 priority encoder. — *ECE-variant (priority encoder asked 5×)*

Q46. **[2M]** *IT 2123 · Dec 2025* Illustrate how an encoder can be used to reduce the number of data lines in a system.

Q47. **[3M]** *ECE 2124 · Jan 2024* Write dataflow Verilog code for a 2:4 decoder. — *ECE-variant (Verilog decoder asked 4×)*

Q48. **[3M]** *ECE 2124 · Dec 2024* Write a dataflow Verilog code for a 2:4 decoder. — *ECE-variant (Verilog decoder asked 4×)*

Q49. **[3M]** *ECE 2124 · Nov/Dec 2024* Write dataflow Verilog code for a 2:4 decoder with active-low enable and active-high output. — *ECE-variant (Verilog decoder asked 4×)*

Q50. **[3M]** *ECE 2153 · Dec 2022* Write a dataflow Verilog code for a 2-to-4 decoder with active-low enable input and active-low outputs. — *ECE-variant (Verilog decoder asked 4×)*

Code converters

Q51. **[5M]** *IT 2123 · Nov/Dec 2023* Design a code converter using NAND gates only to convert a decimal digit represented in 8421 code to a decimal digit represented in Gray code. *(code converters asked 12×)*

Q52. **[5M]** *IT 2123 · Jan 2024* Design a code converter using basic gates only to convert an 8 4 −2 −1 code to a decimal digit represented in 8 4 2 1. *(code converters asked 12×)*

Q53. **[5M]** *ICT 2123 · Nov/Dec 2024* Design a code converter to convert a decimal digit represented in 84-2-1 code to a decimal digit represented in Excess-3 code using a 3-to-8 decoder with active-low output and external gates. *(code converters asked 12×)*

Q54. **[5M]** *ICT 2154 · Jan/Feb 2023* Design a code converter to convert a decimal digit represented in excess-3 to 8 4 −2 −1 code using 3:8 decoders and minimum external gates. *(code converters asked 12×)*

Q55. **[5M]** *ICT 2154 · Dec 2022* Design a code converter to convert a decimal digit represented in 8 4 −2 −1 code to a decimal digit represented in gray code using 3:8 decoders and minimum external gates. *(code converters asked 12×)*

Q56. **[5M]** *ECM 2122 · 2024 Makeup* Design a code converter that converts a decimal digit from 8, 4, −2, −1 code to BCD. *(code converters asked 12×)*

Q57. **[5M]** *ICT 2123 · Dec 2024* Design a 3-bit binary-to-Gray code converter using only basic gates (AND, OR, NOT). Provide the truth table and logic circuit. *(code converters asked 12×)*

Q58. **[5M]** *CSE 2123 · Nov/Dec 2024* Using K-map simplification, design a code converter with the minimum number of logic gates that converts a self-complementing 8-4-2-1 code to the Excess-3 equivalent of BCD. Draw the logic diagram and write the dataflow Verilog code with proper syntax. *(code converters asked 12×)*

Q59. **[4M]** *ECE 2124 · Nov/Dec 2023* Design a combinational circuit for conversion of code 2 4 2 1 to 8 4 2 1 code. — *ECE-variant (code converters asked 12×)*

Q60. **[4M]** *ECE 2124 · Jan 2024* Design a code converter to convert a 3-bit binary to Gray code. — *ECE-variant (code converters asked 12×)*

Q61. **[4M]** *ECE 2124 · Dec 2024* Design a combinational circuit to convert 4-bit Binary to Gray code. — *ECE-variant (code converters asked 12×)*

Q62. **[4M]** *ECE 2124 · Nov/Dec 2024* Implement a 4-bit BCD to Excess-3 code converter circuit using basic gates. — *ECE-variant (code converters asked 12×)*

Q63. **[4M]** *ECE 2124 · Nov 2025* Design a 4-bit Excess-3 to BCD code converter and implement the minimised Boolean expression using NAND gates only. — *ECE-variant (code converters asked 12×)*

Q64. **[4M]** *ECE 2153 · Jan/Feb 2023* Design a code converter that converts an 8421 BCD code into excess-3 code, with four inputs A, B, C, D and four outputs f1, f2, f3, f4. Implement the circuit using a 3:8 decoder with active-low outputs. — *ECE-variant*

Q65. **[3M]** *ECE 2153 · Jan/Feb 2023* Design a combinational circuit to convert the decimal numbers 0, 1, 4, 6, 7 and 8 in a 4-bit Excess-3 code input. — *ECE-variant*

Q66. **[3M]** *ECE 2153 · Jan/Feb 2023* Write structural Verilog code for a 4-bit binary-to-gray code converter using an XOR gate as the basic building block. — *ECE-variant*

Q67. **[3M]** *CSE 2123 · Nov/Dec 2024* Design an optimal circuit for a BCD to seven-segment decoder using logic gates. The seven-segment LED requires a logic high to make a particular LED segment glow.

Arithmetic comparison circuits

Q68. **[3M]** *CSS 2104 · Nov 2025* Design a 3-bit magnitude comparator for two unsigned binary numbers. Derive the final Boolean expressions and draw the equivalent logic circuit diagram. *(magnitude comparator asked 7×)*

Q69. **[3M]** *IT 2123 · Nov 2025* Construct an arithmetic comparison circuit to compare two 2-bit numbers. *(magnitude comparator asked 7×)*

Q70. **[2M]** *ICT 2123 · Dec 2024* Design a 2-bit magnitude comparator using basic gates. *(magnitude comparator asked 7×)*

Q71. **[2M]** *ICT 2123 · Nov/Dec 2024* Design a 1-bit magnitude comparator with cascading inputs using NAND gates only. *(magnitude comparator asked 7×)*

Q72. **[2M]** *IT 2123 · Jan 2024* Design a 1-bit magnitude comparator using basic gates. *(magnitude comparator asked 7×)*

Q73. **[4M]** *ECE 2124 · Dec 2024* Design a 2-bit magnitude comparator using basic logic gates. — *ECE-variant (magnitude comparator asked 7×)*

Q74. **[2M]** *ICT 2154 · Jan/Feb 2023* Design a combinational circuit that compares two 2-bit numbers to check if A ≥ B using logic gates. The circuit output is 1 if the condition is satisfied. *(magnitude comparator asked 7×)*

Q75. **[2M]** *ICT 2154 · Dec 2022* Design a combinational circuit that compares two 4-bit numbers to check if they are equal using logic gates. Output 1 if equal, 0 otherwise.

Q76. **[4M]** *ECE 2124 · Dec 2024* Write a behavioral Verilog code for a 2-bit comparator using an if-else statement. — *ECE-variant*

Q77. **[3M]** *ECE 2124 · Nov/Dec 2023* Write dataflow Verilog code for a given circuit. [figure/table not in text] — *ECE-variant*

Q78. **[3M]** *ECE 2124 · Nov 2025* For the given circuit, write Verilog dataflow modelling considering a delay of 10 ns for the execution of the function. Also draw the timing diagram for inputs and outputs. — *ECE-variant*

Q79. **[3M]** *ECM 2122 · Dec 2025* Explain the Time Division Multiple Access bus arbitration protocol.

Q80. **[5M]** *CSE 2123 · Nov/Dec 2024* Design a digital circuit for a security monitoring system that checks the status of a 4-bit sensor array (A3, A2, A1, A0), counting the number of active sensors (1's in the input) and outputting this count in binary.

Module 4 — Synchronous Sequential Circuits

Flip-flops · Verilog constructs for storage elements · design of synchronous sequential circuits · design of synchronous counters · ripple counters · registers · shift registers · ring and Johnson counters · Verilog constructs for registers and counters

Flip-flops and latches

Q1. **[2M]** *CSE 2123 · Dec 2025* Apply the understanding of sequential circuits to design a basic RS latch using NOR gates. Demonstrate its working with a detailed circuit diagram and truth table. *(RS latch asked 5×)*

Q2. **[2M]** *CSE 2153 / CSE 2123 · Jan 2024* Explain in detail the design of a basic RS latch using NOR gates. *(RS latch asked 5×)*

Q3. **[2M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Demonstrate the SR latch (basic flip-flop circuit) using NAND gates with its function table. *(RS latch asked 5×)*

Q4. **[5M]** *CSE 2153 · Dec 2022* Construct an SR flip-flop using NAND gates. Write its truth table and excitation table. Draw Q_{T+1} waveforms for all combinations of Q_T, S and R, avoiding forbidden conditions. *(RS latch asked 5×)*

Q5. **[5M]** *CSE 2153 · Jan/Feb 2023* Design and explain the clocked NAND gate-based RS flip-flop with the necessary diagram. Derive its characteristic equation. Illustrate the method to overcome the intermediate state in the RS flip-flop by adding only an additional gate. *(RS latch asked 5×)*

Q6. **[3M]** *CSE 2153 · Nov/Dec 2024* Outline the SR flip-flop with a logic diagram using NAND gates, truth table, characteristic equation and excitation table.

Q7. **[2M]** *ICT 2123 · Nov/Dec 2024* Design a D Flip-Flop using an SR Latch of NAND gates. *(flip-flop conversion asked 11×)*

Q8. **[2M]** *IT 2123 · Nov/Dec 2023* Convert the JK flip-flop to a D flip-flop with the necessary steps. *(flip-flop conversion asked 11×)*

Q9. **[2M]** *IT 2123 · Jan 2024* Construct a JK flip-flop using a D flip-flop, a 2-to-1 line multiplexer and inverter only. *(flip-flop conversion asked 11×)*

Q10. **[2M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Construct a JK flip-flop from a D flip-flop. *(flip-flop conversion asked 11×)*

Q11. **[3M]** *ECM 2122 · 2024 Regular* Convert an SR flip-flop to a JK flip-flop. *(flip-flop conversion asked 11×)*

Q12. **[3M]** *ECE 2124 · Nov/Dec 2023* Illustrate how an SR flip-flop can be converted to a JK flip-flop. Also write characteristic equations for the JK flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q13. **[3M]** *ECE 2124 · Jan 2024* Derive the characteristic equation of the SR flip-flop. Convert the SR flip-flop to a D flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q14. **[3M]** *ECE 2124 · Dec 2024* Convert a D flip-flop to a JK flip-flop. Write characteristic equations of the JK flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q15. **[3M]** *ECE 2124 · Nov/Dec 2024* Convert a D flip-flop to a JK flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q16. **[3M]** *ECE 2153 · Jan/Feb 2023* Derive the characteristic equation of the S-R flip-flop. Convert the S-R flip-flop to a J-K flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q17. **[3M]** *ECE 2153 · Dec 2022* Derive the characteristic equation of the P-Q flip-flop given its characteristic table. Convert the P-Q flip-flop to a D flip-flop. — *ECE-variant (flip-flop conversion asked 11×)*

Q18. **[3M]** *CSE 2123 · Dec 2025* Apply the necessary external logic connections to a JK Flip-Flop to convert it into an equivalent T Flip-Flop. Explain the simple modification required on the J and K inputs to achieve the toggle function.

Q19. **[3M]** *ECE 2124 · Nov 2025* Apply the concept of the characteristic table to convert a JK flip-flop into an equivalent D flip-flop. — *ECE-variant*

Q20. **[2M]** *CSE 2123 · Nov 2025* Examine the JK flip-flop with a logic diagram, truth table, characteristic equation and excitation table.

Q21. **[2M]** *ICT 2123 · Dec 2024* Design a G flip-flop according to the given table using a NOR latch. [figure/table not in text]

Q22. **[2M]** *CSS 2104 · (undated)* Write the Verilog code for the K flip-flop whose function table is given below. [figure/table not in text]

Q23. **[2M]** *ICT 2154 · Jan/Feb 2023* Design a T flip-flop using a NOR latch and external gates. *(T flip-flop from latch asked 2×)*

Q24. **[2M]** *ICT 2154 · Dec 2022* Design a T flip-flop using a NAND latch and external gates. *(T flip-flop from latch asked 2×)*

Q25. **[3M]** *CSE 2153 · Jan/Feb 2023* Realize a JK flip-flop using AND and NOR gates.

Q26. **[3M]** *CSE 2123 · Nov 2025* Determine the phenomenon of "race-around" that can occur in a simple edge-triggered JK Flip-Flop. Explain how the architecture of a Master/Slave Flip-Flop prevents this problem by isolating the input from the output during the critical clock transition. *(race-around / master-slave asked 3×)*

Q27. **[2M]** *CSE 2153 / CSE 2123 · Jan 2024* Explain how the restriction on the pulse width in the case of the JK flip-flop can be solved by using a master/slave JK flip-flop. *(race-around / master-slave asked 3×)*

Q28. **[3M]** *ECE 2124 · (undated)* What is the race-around condition in JK flip-flops? Explain how a Master-Slave flip-flop can solve the race-around condition with a suitable logic diagram. — *ECE-variant (race-around / master-slave asked 3×)*

Q29. **[4M]** *CSE 2153 / CSE 2123 · Jan 2024* Build the D-type positive edge-triggered flip-flop using NAND gates. Explain its operation with logic diagrams for all cases of D when the clock is low. Predict the outputs of the flip-flop for each case assuming the flip-flop was cleared previously.

Q30. **[4M]** *IT 2123 · Dec 2025* Apply Verilog constructs to model a D flip-flop with synchronous reset. *(Verilog flip-flop asked 3×)*

Q31. **[3M]** *ECE 2124 · Jan 2024* Write the sequential Verilog code for a positive edge-triggered JK flip-flop. — *ECE-variant (Verilog flip-flop asked 3×)*

Q32. **[3M]** *ECE 2153 · Dec 2022* Write the sequential Verilog code for a positive edge-triggered J-K flip-flop. — *ECE-variant (Verilog flip-flop asked 3×)*

Q33. **[2M]** *CSE 2153 · Dec 2022* Identify the memory element used in sequential circuits and draw the logical diagram of a single-input memory element.

Design of synchronous sequential circuits

Q34. **[5M]** *CSS 2104 · (undated)* Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If E = 0, the circuit remains in the same state regardless of F. When E = 1 and F = 1, the circuit goes through the state transitions 00 → 01 → 10 → 11 → 00 and repeats. When E = 1 and F = 0, the circuit goes through 00 → 11 → 10 → 01 → 00 and repeats. *(JK/T/D state-diagram design asked 8×)*

Q35. **[5M]** *CSE 2123 · Dec 2025* Design a sequential circuit with two JK flip-flops A and B and two inputs E and F, with the same specification as above. *(JK/T/D state-diagram design asked 8×)*

Q36. **[5M]** *CSS 2104 · Nov 2025* Analyze the given state diagram and design the corresponding synchronous sequential circuit using JK flip-flops. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q37. **[5M]** *CSE 2123 · Nov 2025* Apply the complete synchronous sequential circuit design procedure to realize the functionality described by a given state diagram. The final circuit must be implemented using JK Flip-Flops. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q38. **[5M]** *CSE 2123 · Dec 2024* Design a synchronous sequential circuit for the following state diagram using a JK flip-flop. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q39. **[5M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* For the following state diagram, derive the expressions required for the synchronous sequential circuit design using JK flip-flops. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q40. **[5M]** *CSE 2153 / CSE 2123 · Jan 2024* For the following state diagram, derive the expressions required for the synchronous sequential circuit design using T flip-flops. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q41. **[5M]** *CSE 2123 · Nov/Dec 2024* Reduce the number of states in the following state diagram and design a synchronous sequential circuit for the reduced state diagram using T-flip-flops. [figure/table not in text] *(JK/T/D state-diagram design asked 8×)*

Q42. **[4M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Draw the state table and design the synchronous sequential circuit for the following state diagram using JK FF. [figure/table not in text]

Q43. **[5M]** *ECM 2122 · Dec 2025* Design a Mealy synchronous sequential circuit using D flip-flops with a serial input x and output z that produces z = 1 whenever the input completes the sequence 1011. Overlapping is allowed. *(sequence detector asked 13×)*

Q44. **[5M]** *ECM 2122 · Nov 2025* Design a Mealy synchronous sequential circuit using D flip-flops that will function as prescribed by the given state diagram. [figure/table not in text] *(sequence detector asked 13×)*

Q45. **[5M]** *ICT 2123 · Dec 2024* Analyze and design a sequential circuit based on the Mealy model that uses D flip-flops to detect the sequence "0101". The circuit should produce an active-high output only when the sequence is detected. Ensure the design allows overlapping. *(sequence detector asked 13×)*

Q46. **[5M]** *ICT 2123 · Nov/Dec 2024* Design a Mealy machine that detects the sequence "1101". (a) How many states are required? (b) Construct the state transition table. (c) Given an input sequence of "1101101", determine the output sequence. *(sequence detector asked 13×)*

Q47. **[5M]** *IT 2123 · Nov/Dec 2023* Design a sequence detector to detect three or more consecutive 1's in a sequence of bits using the Mealy model. *(sequence detector asked 13×)*

Q48. **[5M]** *IT 2123 · Jan 2024* Design the circuit diagram for a sequence detector to detect the sequence 011 using the Mealy model. *(sequence detector asked 13×)*

Q49. **[5M]** *ICT 2154 · Jan/Feb 2023* Design a sequence detector using the Moore model with one input Y and one output Z. Z is HIGH whenever the sequence "101" is detected, otherwise LOW. Overlapping is allowed. Implement using JK flip-flops and the minimum number of external gates. *(sequence detector asked 13×)*

Q50. **[5M]** *ICT 2154 · Dec 2022* Design a sequence detector using the Mealy model with one input Y and one output Z. Z is HIGH whenever the sequence "1001" is detected. Overlapping is allowed. Implement using JK flip-flops and minimum external gates. *(sequence detector asked 13×)*

Q51. **[4M]** *ECE 2124 · Nov/Dec 2023* Design a sequence detector to detect 1010 with overlapping using the Moore model with a positive edge-triggered D flip-flop. — *ECE-variant (sequence detector asked 13×)*

Q52. **[4M]** *ECE 2124 · Jan 2024* Design a sequence detector to detect 101 with overlapping using the Mealy model with a positive edge-triggered D flip-flop. — *ECE-variant (sequence detector asked 13×)*

Q53. **[4M]** *ECE 2124 · (undated)* Design a sequential circuit using a Mealy machine that can detect the overlapping sequence 1010. Use positive edge-triggered D flip-flops. — *ECE-variant (sequence detector asked 13×)*

Q54. **[4M]** *ECE 2124 · Nov 2025* Design a sequential circuit using a Mealy machine that can detect the non-overlapping sequence 0110. Use positive edge-triggered D flip-flops. — *ECE-variant (sequence detector asked 13×)*

Q55. **[4M]** *ECE 2124 · Dec 2024* Design a sequence detector to detect the overlapping sequence 0101 using the Mealy model with a positive edge-triggered D flip-flop. — *ECE-variant (sequence detector asked 13×)*

Q56. **[4M]** *ECE 2124 · Nov/Dec 2024* Design a sequential circuit using a Mealy machine that can detect the overlapping sequence 1001 using negative edge-triggered D flip-flops. — *ECE-variant (sequence detector asked 13×)*

Q57. **[3M]** *ECE 2153 · Dec 2022* Design a Mealy overlapping sequence detector that detects the sequence 101 using a D flip-flop and necessary gates. — *ECE-variant*

Q58. **[4M]** *ECE 2153 · Dec 2022* Implement a 1111 overlapping sequence detector using the Mealy model and draw the circuit diagram using D flip-flops and necessary logic gates. — *ECE-variant*

Q59. **[3M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Analyze the following state diagram and perform the state reduction. [figure/table not in text] *(state reduction asked 4×)*

Q60. **[3M]** *CSE 2153 / CSE 2123 · Jan 2024* Analyze the following state table and perform the state reduction. Develop the state diagram for the reduced state table. [figure/table not in text] *(state reduction asked 4×)*

Q61. **[5M]** *CSE 2153 · Jan/Feb 2023* Reduce the number of states in the following state table and tabulate the reduced state table. Compare the output sequence for the given table and the reduced state table using the input sequence starting from state a, and input sequence 110010011. [figure/table not in text] *(state reduction asked 4×)*

Q62. **[4M]** *ECM 2122 · Dec 2025* Analyze the sequential circuit shown below and obtain its state table and state diagram. [figure/table not in text] *(circuit analysis asked 2×)*

Q63. **[4M]** *ECM 2122 · Nov 2025* Analyze the synchronous sequential circuit shown below and obtain its state table and state diagram. [figure/table not in text] *(circuit analysis asked 2×)*

Q64. **[2M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Distinguish between Moore and Mealy models of sequential circuit design.

Q65. **[2M]** *ICT 2123 · Dec 2024* Analyze the state diagram shown and obtain the excitation table using SR flip-flops. [figure/table not in text]

Q66. **[2M]** *CSE 2153 / CSE 2123 · Jan 2024* Develop a state diagram for the FSM meeting this specification: the circuit has one input w and one output z; z should equal 1 in the same clock cycle when the second occurrence of w = 1 is detected, otherwise 0. What output z is produced for the given input sequence? *(FSM design asked 4×)*

Q67. **[3M]** *ECM 2122 · 2024 Makeup* Design an FSM for a two-floor elevator (ground and first floor). One button controls the elevator with values Up or Down. Two lights indicate the current floor: Red for Ground, Green for First. At each time step the controller checks the current floor and input, and changes floors and lights accordingly. *(FSM design asked 4×)*

Q68. **[2M]** *ECM 2122 · 2024 Makeup* Draw an FSM diagram for a serial binary input which is divisible by 4. *(FSM design asked 4×)*

Q69. **[2M]** *ECM 2122 · 2024 Regular* Draw an FSM diagram for a serial binary input which is divisible by 5. *(FSM design asked 4×)*

Q70. **[5M]** *CSE 2123 · Dec 2024* Design a traffic light controller for a four-way intersection allowing North-South (N-S) and East-West (E-W) traffic, where only one direction can have a green light at a time. North-South has higher priority. SENSOR_NS is HIGH if there is traffic in the N-S direction; SENSOR_EW is HIGH for E-W. GREEN_NS and GREEN_EW are output signals controlling the green lights.

Q71. **[5M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* It is required to develop a system to swap the contents of two n-bit registers, assuming a temporary register is available. Adopt the state table approach to design the control circuit, which has two inputs clock and w. The swapping process is initiated when w = 1, and on completion the control circuit asserts a done signal.

Q72. **[5M]** *CSE 2153 / CSE 2123 · Jan 2024* Design a control unit of a simple processor to perform the operations mentioned in the table. (a) Develop the timing signals for the operations. (b) Tabulate the necessary control signals that need to be asserted. (c) Derive the expressions for all the control signals. [figure/table not in text]

Synchronous counters

Q73. **[3M]** *CSS 2104 · (undated)* Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. *(synchronous counter design asked 14×)*

Q74. **[5M]** *ICT 2123 · Dec 2024* Design a 4-bit synchronous binary counter. Draw the circuit using JK flip-flops and explain how the flip-flops are connected to count in binary from 0000 to 1111 in synchronisation with the clock. *(synchronous counter design asked 14×)*

Q75. **[5M]** *CSE 2153 · Jan/Feb 2023* Construct a 4-bit synchronous binary up-counter using JK flip-flops. Draw the waveform for the same. *(synchronous counter design asked 14×)*

Q76. **[3M]** *IT 2123 · Nov 2025* Design a 4-bit synchronous counter using D flip-flops. *(synchronous counter design asked 14×)*

Q77. **[5M]** *ICT 2123 · Nov/Dec 2024* Design a 4-bit synchronous counter to count according to the given sequence using D flip-flops and minimum external gates. All undefined states should go to state 0. [figure/table not in text] *(synchronous counter design asked 14×)*

Q78. **[5M]** *ECM 2122 · Dec 2025* Design a 3-bit self-correcting synchronous counter using JK flip-flops that follows the sequence 7-6-4-3 repeatedly, ensuring that any unused states automatically go to state 7 on the next clock pulse. *(synchronous counter design asked 14×)*

Q79. **[5M]** *ECM 2122 · Nov 2025* Design a 3-bit self-correcting synchronous counter using JK flip-flops that follows the sequence 0-1-2-4 repeatedly, with unused states going to state 0 on the next clock pulse. *(synchronous counter design asked 14×)*

Q80. **[5M]** *ECM 2122 · 2024 Makeup* Design a synchronous counter with the sequence 0, 1, 3, 7, 6, 4, 0 using JK flip-flops (assume unused states as don't care). *(synchronous counter design asked 14×)*

Q81. **[5M]** *CSE 2123 · Dec 2025* Illustrate the synchronous counter with the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6. Use JK flip-flops. Check if it is self-correcting. *(synchronous counter design asked 14×)*

Q82. **[5M]** *CSE 2153 / CSE 2123 · Jan 2024* Design the synchronous counter with the repeated binary sequence 0, 1, 2, 3, 4, 5, 6 using JK flip-flops. Check if it is self-correcting. *(synchronous counter design asked 14×)*

Q83. **[5M]** *CSE 2153 · Dec 2022* Develop the expressions for the flip-flop inputs in the design of a counter with the repeated binary sequence 0, 1, 2, 4, 5, 6, 7. Use JK flip-flops and treat unused states as don't care. Analyze whether the counter is self-correcting. *(synchronous counter design asked 14×)*

Q84. **[5M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Design the synchronous counter with the repeated binary sequence 0, 1, 2, 4, 6 using D flip-flops. Check if it is self-correcting. *(synchronous counter design asked 14×)*

Q85. **[3M]** *CSE 2123 · Dec 2025* Design a three-bit counter that counts in the sequence 0, 1, 2, 4, 5, 6, 0. *(synchronous counter design asked 14×)*

Q86. **[3M]** *IT 2123 · Jan 2024* Design a synchronous counter to generate the sequence 0, 3, 5, 6, 0, ... using JK flip-flops and external gates. *(synchronous counter design asked 14×)*

Q87. **[5M]** *IT 2123 · Jan 2024* Devise a counter circuit employing D flip-flops and minimum external gates to generate the recurrent binary sequence 0, 1, 2, 4, 6. Include the necessary connections (clock and clear inputs), ensuring the counter resets to 0 after reaching 6. Illustrate with a state diagram, excitation table and logic diagram. The undefined state should go to state 0.

Q88. **[5M]** *ECM 2122 · 2024 Regular* Design a 3-bit synchronous down counter using JK flip-flops. *(up/down counters asked 5×)*

Q89. **[3M]** *CSE 2123 · Dec 2024* Design a synchronous 3-bit up-down counter using JK flip-flops. Draw an appropriate sequence diagram, excitation table and logic diagram. *(up/down counters asked 5×)*

Q90. **[5M]** *IT 2123 · Nov/Dec 2023* Design a Synchronous DOWN counter that follows a 3-bit Gray code sequence. Illustrate the counter's operation through a state diagram, excitation table and logic diagram. Use JK flip-flops and external gates. *(up/down counters asked 5×)*

Q91. **[3M]** *CSE 2123 · Nov/Dec 2024* Realize a synchronous 3-bit Gray code up-counter using T flip-flops with an appropriate sequence diagram, excitation table and logic diagram. *(up/down counters asked 5×)*

Q92. **[3M]** *ECE 2124 · Dec 2024* Design a synchronous 3-bit Gray code counter using positive edge-triggered T flip-flops. — *ECE-variant (up/down counters asked 5×)*

Q93. **[3M]** *CSE 2123 · Dec 2024* Design a 3-bit synchronous counter using D flip-flops that counts even states only and repeats the sequence after reaching the maximum even number, starting from state 0. Assume unused states are don't cares.

Q94. **[5M]** *ICT 2154 · Dec 2022* Design and implement a MOD-5 synchronous counter using JK flip-flops and draw its timing diagram. *(MOD-N counters asked 4×)*

Q95. **[5M]** *ICT 2154 · Jan/Feb 2023* Design a synchronous divide-by-10 counter using D flip-flops. *(MOD-N counters asked 4×)*

Q96. **[4M]** *ECE 2124 · (undated)* Design a 3-bit synchronous Mod-5 counter using T flip-flops. — *ECE-variant (MOD-N counters asked 4×)*

Q97. **[3M]** *ECE 2124 · Jan 2024* Design a synchronous mod-8 up counter using T flip-flops. — *ECE-variant (MOD-N counters asked 4×)*

Q98. **[4M]** *ECE 2124 · Nov 2025* Design a synchronous counter that follows the counting sequence 0, 1, 3, 4, 2, and then repeats. Ensure all unused states return to 0. Use positive edge-triggered T flip-flops. — *ECE-variant*

Q99. **[3M]** *ECE 2124 · Nov/Dec 2024* Design a self-starting counter using positive edge-triggered D flip-flops to generate the sequence 1, 2, 5, 7 and repeat. Avoid lock-out by forcing unused states 0 to 1, 3 and 4 to 5, and 6 to 7. — *ECE-variant*

Q100. **[3M]** *ECE 2153 · Dec 2022* Construct a 2-bit synchronous up counter using T flip-flops. — *ECE-variant*

Q101. **[3M]** *ECE 2153 · Jan/Feb 2023* Construct a 2-bit synchronous down counter using T flip-flops. — *ECE-variant*

Q102. **[2M]** *ICT 2123 · Nov/Dec 2024* Define a MOD-N counter. How does the value of N affect the design of the synchronous counter?

Q103. **[2M]** *CSE 2123 · Dec 2024* Determine the number of unused states and MOD of the given counter. [figure/table not in text]

Ripple (asynchronous) counters

Q104. **[3M]** *ECM 2122 · Dec 2025* Construct a Mod-10 ripple up-counter using negative edge-triggered T flip-flops and draw the output timing diagrams. *(ripple counter asked 12×)*

Q105. **[3M]** *ECM 2122 · Nov 2025* Implement a mod-12 ripple up-counter using negative edge-triggered JK flip-flops and draw the output timing diagrams. *(ripple counter asked 12×)*

Q106. **[3M]** *ICT 2123 · Dec 2024* Design a 3-bit asynchronous counter using JK flip-flops. Provide the circuit diagram and explain how state transitions occur with clock inputs. *(ripple counter asked 12×)*

Q107. **[3M]** *CSS 2104 · Nov 2025* Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK flip-flops and explain its working. *(ripple counter asked 12×)*

Q108. **[5M]** *ECM 2122 · 2024 Makeup* Design an asynchronous 3-bit up counter using negative edge-triggered JK/T flip-flops and write Verilog code for the same in dataflow modelling format. *(ripple counter asked 12×)*

Q109. **[5M]** *ECM 2122 · 2024 Regular* Design an asynchronous 3-bit down counter using negative edge-triggered JK/T flip-flops and write Verilog code for the same in dataflow modelling format. *(ripple counter asked 12×)*

Q110. **[2M]** *IT 2123 · Nov/Dec 2023* Design an Asynchronous DOWN counter to count from 6 to 2 continuously using negative edge-triggered T flip-flops and minimum external gates. *(ripple counter asked 12×)*

Q111. **[2M]** *IT 2123 · Jan 2024* Design an Asynchronous UP counter to count from 2 to 6 continuously using positive edge-triggered JK flip-flops and minimum external gates. *(ripple counter asked 12×)*

Q112. **[3M]** *ICT 2154 · Jan/Feb 2023* Design an asynchronous UP counter to count from 2 to 6 using positive edge-triggered D flip-flops and minimum external gates. *(ripple counter asked 12×)*

Q113. **[3M]** *ICT 2154 · Dec 2022* Design an asynchronous DOWN counter to count from 5 to 2 using positive edge-triggered D flip-flops and minimum external gates. *(ripple counter asked 12×)*

Q114. **[3M]** *ECE 2124 · Nov/Dec 2023* Design a 3-bit asynchronous down counter using T flip-flops. — *ECE-variant (ripple counter asked 12×)*

Q115. **[3M]** *ECE 2124 · Jan 2024* Design a Mod-10 ripple up counter using T flip-flops. — *ECE-variant (ripple counter asked 12×)*

Q116. **[3M]** *ECE 2124 · Dec 2024* Design a mod-6 ripple up counter using negative edge-triggered T flip-flops with active-high clear. — *ECE-variant*

Q117. **[3M]** *ECE 2124 · Nov 2025* Design an asynchronous decade counter using positive edge-triggered JK flip-flops. — *ECE-variant*

Q118. **[3M]** *ECE 2153 · Jan/Feb 2023* Design a mod-6 asynchronous up-counter using T flip-flops. — *ECE-variant*

Q119. **[3M]** *ECE 2124 · (undated)* Design a 4-bit asynchronous up-down counter using positive edge-triggered T flip-flops. — *ECE-variant*

Q120. **[3M]** *ECE 2124 · Nov/Dec 2024* Design an asynchronous up-counter using positive edge-triggered JK flip-flops to divide the input frequency by 8, with waveforms for the clock and flip-flop outputs. — *ECE-variant*

Q121. **[2M]** *IT 2123 · Jan 2024* Draw the circuit diagram of a 2-bit Up-Down Asynchronous counter using positive-edge D flip-flops and external gates.

Q122. **[2M]** *CSE 2153 · Dec 2022* Design a 3-bit ripple up-counter using positive edge-triggered T flip-flops.

Q123. **[2M]** *CSE 2123 · Nov/Dec 2024* Design a MOD-6 ripple counter using T flip-flops and explain its working using a timing diagram.

Q124. **[2M]** *ICT 2123 · Dec 2024* Using your understanding of asynchronous counters and SR flip-flops, design a frequency divider circuit that reduces the system clock frequency by a factor of 1/6.

Q125. **[3M]** *IT 2123 · Dec 2025* Differentiate between asynchronous counter and synchronous counter in terms of timing. *(sync vs async counters asked 3×)*

Q126. **[4M]** *IT 2123 · Nov 2025* Differentiate between ripple counter and synchronous counter in terms of timing. *(sync vs async counters asked 3×)*

Q127. **[3M]** *CSE 2123 · Nov 2025* Determine the key structural difference between a Synchronous Counter and an Asynchronous Counter. Justify why the synchronous approach, where all flip-flops are clocked simultaneously, is essential for high-speed operation. *(sync vs async counters asked 3×)*

Q128. **[2M]** *ICT 2154 · Jan/Feb 2023* Using synchronous hexadecimal counter ICs and external gates, design a 2-digit hexadecimal down counter which counts from 94H to 35H and repeats.

Q129. **[2M]** *ICT 2154 · Dec 2022* Using asynchronous hexadecimal counter ICs and external gates, design a 2-digit hexadecimal counter which counts from 00 to 89H and repeats.

Q130. **[2M]** *IT 2123 · Nov/Dec 2023* All four flip-flops' inputs are made high in the figure below. Draw the transition diagram and find the signal's frequency at the output Y. [figure/table not in text]

Registers and shift registers

Q131. **[5M]** *CSE 2123 · Nov 2025* Design and illustrate a 4-bit shift register with parallel load that operates according to the given function table. [figure/table not in text] *(shift register with parallel load asked 3×)*

Q132. **[5M]** *CSE 2123 · Nov/Dec 2024* Design and illustrate a 4-bit shift register with parallel load that operates according to the given function table. [figure/table not in text] *(shift register with parallel load asked 3×)*

Q133. **[2M]** *CSE 2153 · Jan/Feb 2023* Design a 5-bit shift register with parallel load that operates according to the function table shown. [figure/table not in text] *(shift register with parallel load asked 3×)*

Q134. **[3M]** *ECE 2124 · Nov/Dec 2023* Design a 4-bit universal shift register which can handle the following functionalities. [figure/table not in text] — *ECE-variant (universal shift register asked 2×)*

Q135. **[4M]** *ECE 2153 · Jan/Feb 2023* Illustrate the working of a 4-bit universal shift register with the help of a neat diagram as per the given table. — *ECE-variant (universal shift register asked 2×)*

Q136. **[3M]** *CSE 2123 · Dec 2025* Determine the operation of a Serial-In, Parallel-Out (SIPO) Shift Register. Explain the specific steps (in terms of clock cycles) required to load an 8-bit data word into the register and how the data is then accessed at the output terminals.

Q137. **[4M]** *ECE 2124 · (undated)* Write a behavioral Verilog code for a 4-bit Serial-In Serial-Out (SISO) shift register using positive edge-triggered D flip-flops. — *ECE-variant*

Q138. **[2M]** *IT 2256 · May/Jun 2023* The 4-bit shifter has four data inputs A0 through A3 and four data outputs H0 through H3. There are two serial inputs, one for shift left (IL) and the other for shift right (IR). Design according to the given requirements. [figure/table not in text]

Q139. **[3M]** *ICT 2123 · Dec 2024* Design a 4×4 combinational shifter using MUX and tristate buffer.

Q140. **[2M]** *IT 2256 · Jul 2023* Design a 4-bit general purpose register according to the given truth table. [figure/table not in text]

Q141. **[5M]** *CSE 2123 · Dec 2025* Draw the standard symbol and the truth table for a Tri-state Driver. Apply the Tri-state Driver to explain how the outputs of two registers R1 and R2 are safely connected to a common bus line with the help of a diagram. *(tri-state drivers asked 4×)*

Q142. **[2M]** *CSE 2153 · Jan/Feb 2023* Demonstrate the working of tri-state drivers. *(tri-state drivers asked 4×)*

Q143. **[3M]** *CSE 2153 · Jan/Feb 2023* It is required to design a digital system with the operations shown in the table. There are four registers R0 to R3 connected to the bus through tri-state drivers. External data can be loaded into the registers from the Data input, connected to the bus using tri-state drivers enabled by an External signal. [figure/table not in text] *(tri-state drivers asked 4×)*

Q144. **[5M]** *CSE 2153 · Dec 2022* It is required to design a digital system with the operations shown in the table, with four registers R0 to R3 connected to the bus through tri-state drivers. [figure/table not in text] *(tri-state drivers asked 4×)*

Q145. **[2M]** *CSE 2153 · Dec 2022* Illustrate the data transfer between registers via an interconnection network in a computer system.

Ring and Johnson counters

Q146. **[3M]** *IT 2123 · Dec 2025* Design a 4-bit Johnson Counter and discuss its applications. *(Johnson counter asked 8×)*

Q147. **[3M]** *CSE 2123 · Nov 2025* Design a 3-bit Johnson counter by implementing the following steps: (a) draw the state-transition diagram (b) construct a state table using T flip-flops (c) design the counter. *(Johnson counter asked 8×)*

Q148. **[3M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Design a 3-bit Johnson counter: (a) draw the state-transition diagram (b) construct the state table using T flip-flops (c) design the counter. *(Johnson counter asked 8×)*

Q149. **[2M]** *CSE 2153 · Dec 2022* Construct a Johnson counter which generates 6 timing signals. Write the count sequence and the AND gates required for the output. *(Johnson counter asked 8×)*

Q150. **[3M]** *CSE 2153 / CSE 2123 · Jan 2024* Construct a Johnson counter which generates 8 timing signals. Write the count sequence and AND gate expressions required for decoding the output. *(Johnson counter asked 8×)*

Q151. **[3M]** *ICT 2123 · Nov/Dec 2024* Using the Johnson counter, generate the sequence 010110. Use JK flip-flops and minimum external gates. *(Johnson counter asked 8×)*

Q152. **[3M]** *ICT 2154 · Dec 2022* Design a sequence generator using a Johnson counter (shift register) to generate the sequence 1101. *(Johnson counter asked 8×)*

Q153. **[2M]** *CSE 2123 · Nov 2025* Determine the number of unique states generated by a k = 5 bit Johnson Counter. Write the general formula relating the number of states to the number of flip-flops k. *(Johnson counter asked 8×)*

Q154. **[2M]** *CSS 2104 · Nov 2025* Write behavioral Verilog code that represents an eight-bit ring counter. *(ring counter asked 6×)*

Q155. **[2M]** *CSE 2123 · Nov/Dec 2024* Brief the ring counter and write Verilog code that represents an eight-bit ring counter. *(ring counter asked 6×)*

Q156. **[3M]** *IT 2123 · Nov 2025* Develop a Verilog code for a 4-bit Ring Counter. *(ring counter asked 6×)*

Q157. **[2M]** *CSE 2153 / CSE 2123 · Nov/Dec 2023* Design a 4-bit ring counter using D flip-flops. *(ring counter asked 6×)*

Q158. **[2M]** *CSE 2153 / CSE 2123 · Jan 2024* Construct a 5-bit ring counter using D flip-flops. *(ring counter asked 6×)*

Q159. **[2M]** *CSE 2123 · Dec 2025* Show the relationship between the number of states N generated by a simple ring counter and the number of flip-flops k used. *(ring counter asked 6×)*

Q160. **[3M]** *ICT 2154 · Jan/Feb 2023* Design a sequence generator to generate the sequence 1101011 by ring counter using JK flip-flops and external gates.

Q161. **[3M]** *IT 2123 · Nov/Dec 2023* Design a sequence generator using a synchronous counter with states 0 → 3 → 2 → 1 → 0 that generates the sequence '1100'. Use T flip-flops and minimum external gates.

Verilog for registers and counters

Q162. **[4M]** *ECM 2122 · Dec 2025* A mod-6 asynchronous up-counter using negative edge-triggered T flip-flops, which resets itself if any undesired state occurs, needs to be designed. Write the Verilog code to implement the same.

Q163. **[4M]** *ECM 2122 · Nov 2025* Write the Verilog code in behavioral modeling to implement a 3-bit synchronous binary counter using negative edge-triggered T flip-flops.

Q164. **[2M]** *CSE 2123 · Dec 2025* Apply the design principles of an Asynchronous (Ripple) Counter and the operational constraints of a Positive Edge-Triggered T Flip-Flop to implement a 4-bit UP counter in Verilog.

Q165. **[2M]** *CSE 2123 · Dec 2024* Write the Verilog behavioral description of a four-bit up-down counter with parallel load using three control inputs for the three operations Load, Up and Down. The order of precedence is Load, Up, Down.

Q166. **[3M]** *CSE 2123 · Nov/Dec 2024* Design a three-bit up/down counter using D flip-flops in Verilog. Include a control input called Up/Down: if Up/Down = 0 the circuit behaves as an up-counter; if 1, as a down-counter.

Module 5 — Instruction Set Architecture

Memory locations and addresses · memory operations · instructions and instruction sequencing · addressing modes · CISC instruction sets · RISC and CISC styles · example programs

CISC and RISC example programs

Q1. **[4M]** *CSS 2104 · (undated)* In a signal processing application, two equal-length vectors A and B represent input and weight values of a digital filter, each containing N elements. Develop a CISC-style assembly program to compute the dot product of vectors A and B and store the result in memory location DOTPROD. Describe the sequence of operations required. *(dot product program asked 5×)*

Q2. **[4M]** *CSS 2104 · Nov 2025* Analyze the architectural differences between RISC and CISC processors. Write a CISC-style program to compute the dot product of two given vectors. *(dot product program asked 5×)*

Q3. **[4M]** *CSE 2121 · Nov 2025* In a robotics laboratory, a CISC-based processor is programmed to compute the dot product of two sensor data arrays SENSOR_A and SENSOR_B, each containing N elements. The processor loads base addresses and initializes counter COUNT with N; each iteration loads one element from each array, multiplies them, and adds the product. Describe the sequence of operations. *(dot product program asked 5×)*

Q4. **[5M]** *CSE 2121 · Nov/Dec 2024* In a robotics lab, a CISC-based processor calculates the dot product of two sensor data arrays SENSOR_A and SENSOR_B, each with N elements, storing the result in DOT_PRODUCT. Describe the sequence of operations. *(dot product program asked 5×)*

Q5. **[3M]** *CSE 2151 · Jan/Feb 2023* Assume array A is stored in consecutive word locations starting from address A and array B starting from address B. Write a CISC-style program to find the product of the corresponding elements of two arrays and store it in a third array C starting at address C. Assume uniform-sized arrays with the size stored in memory location SIZE, byte-addressable memory and word length of two bytes. *(dot product program asked 5×)*

Q6. **[3M]** *CSE 2151 · Dec 2022* Assume array A is stored in consecutive word locations starting from address A and array B starting from address B. Write a RISC-style program to find the product of the corresponding elements and store it in array C starting at address C.

Q7. **[4M]** *CSE 2121 · Jan 2026* A list of 32-bit positive integers is stored in memory from the location named LIST. The word at address ELE is to hold the value of the smallest number after it has been found. Assuming the number of entries is at memory location N, write a RISC-style program to find the smallest number in the list. *(smallest-in-list program asked 2×)*

Q8. **[5M]** *CSE 2121 / CSE 2151 · Jan 2024* A list of 32-bit positive integers is stored in memory from the location named LIST, with the smallest to be stored at address ELE and the count at location N. Write a RISC-style program to find the smallest number in the list. *(smallest-in-list program asked 2×)*

Q9. **[5M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Assume a list of non-repetitive integers is present in byte-addressable memory starting from location 1000, with the number of elements at location N. Using Indexed Addressing Mode, write a RISC-style program that searches for an element in this list.

Q10. **[2M]** *CSE 2121 · Dec 2024* For the RISC architecture, list 6 addressing modes with definitions and explain (i) Index addressing mode (ii) Indirect addressing mode. Also: a list of N numbers is stored in memory — write an assembly program to multiply the list of N and store the product back into memory using RISC instructions, and explain the importance of each conditional flag code during instruction execution.

Addressing modes

Q11. **[4M]** *ECM 2122 · Dec 2025* Define the following addressing modes with an example for each: (i) register-indirect (ii) displacement (iii) autodecrement (iv) implied (implicit). *(addressing modes asked 5×)*

Q12. **[4M]** *ECM 2122 · Nov 2025* Define the following addressing modes with an example for each: (i) register-indirect (ii) autoincrement (iii) indexed (displacement) (iv) immediate. *(addressing modes asked 5×)*

Q13. **[3M]** *ECM 2122 · 2024 Makeup* Explain the following addressing modes with examples: (1) Displacement (2) Indirect (3) Register direct. *(addressing modes asked 5×)*

Q14. **[3M]** *ECM 2122 · 2024 Regular* Explain the following addressing modes with examples: (1) Auto increment (2) Implied (3) Register indirect. *(addressing modes asked 5×)*

Q15. **[5M]** *IT 2256 · Jul 2023* Explain the direct, register indirect and immediate addressing modes. Categorize the given instructions according to the source addressing mode used. *(addressing modes asked 5×)*

Q16. **[2M]** *IT 2123 · Nov 2025* Apply the concept of addressing modes to write an instruction using immediate addressing.

Q17. **[3M]** *CSS 2104 · Nov 2025* Explain the concept of memory addresses in an Instruction Set Architecture. Also discuss the difference between byte addressing and word addressing. *(memory addressing asked 2×)*

Q18. **[5M]** *IT 2256 · 2024* Present Big endian and Little endian methods of byte addressing with examples. *(memory addressing asked 2×)*

Instruction formats

Q19. **[4M]** *ECM 2122 · Dec 2025* Express the given statement in three, two and one instruction addressing formats: D = A + B×C. *(instruction address formats asked 6×)*

Q20. **[4M]** *ECM 2122 · Nov 2025* Express the given statement in three, two and one instruction addressing formats: D = (A+B) × (C+D). *(instruction address formats asked 6×)*

Q21. **[2M]** *ECM 2122 · 2024 Makeup* Express the given statement in three instruction format: D = A + BC. (instruction address formats asked 6×)*

Q22. **[2M]** *ECM 2122 · 2024 Regular* Express the given statement in two instruction format: D = A + BC. (instruction address formats asked 6×)*

Q23. **[5M]** *ECE 2152 · Jan/Feb 2023* For the expression X = (A−B/C)/(D + E/F + G*H), write the program using 3-address, 2-address, 1-address and 0-address instructions. Assume source and destination operands are memory locations.* — ECE-variant (instruction address formats asked 6×)*

Q24. **[4M]** *ECE 2152 · Nov/Dec 2023* Write a program using 3, 2, 1 and 0 address instructions for Y = (A−B)*(C+D)/E. Assume source and destination operands are memory locations. Write comments for each step.* — ECE-variant (instruction address formats asked 6×)*

Q25. **[5M]** *ECE 2152 · Dec 2022* For the expression X = (A−B/C)/(D+E*F − G/H), write the program for register-based, accumulator-based and stack-based machines.* — ECE-variant*

Q26. **[3M]** *ECE 2152 · Jan/Feb 2023* In a computer instruction format, the instruction length is 12 bits and the address field is 4 bits. The instruction set has 14 two-address and 50 zero-address instructions. Using the expanding opcode technique, compute the number of one-address instructions. — *ECE-variant (expanding opcode asked 3×)*

Q27. **[3M]** *ECE 2152 · Dec 2022* With 12-bit instruction length and 4-bit address field, verify the possibility of having fourteen 2-address, thirty 1-address and twenty-four 0-address instructions. — *ECE-variant (expanding opcode asked 3×)*

Q28. **[3M]** *ECE 2152 · Nov/Dec 2023* With 11-bit instruction length and 4-bit address field, verify the possibility of having five 2-address, forty-five 1-address and thirty-two 0-address instructions. — *ECE-variant (expanding opcode asked 3×)*

RISC vs CISC and processor organization

Q29. **[5M]** *IT 2123 · Nov 2025* Compare the characteristics of CISC and RISC architectures with examples. *(RISC vs CISC asked 3×)*

Q30. **[3M]** *ECE 2152 · Dec 2022* What is CISC architecture? Explain its characteristic features. — *ECE-variant (RISC vs CISC asked 3×)*

Q31. **[4M]** *ECE 2152 · Nov/Dec 2023* Compare the architectural features of CISC, RISC and VLIW architectures. — *ECE-variant (RISC vs CISC asked 3×)*

Q32. **[2M]** *ECM 2122 · Dec 2025* Explain the operation of accumulator-based computer organization. *(processor organization asked 5×)*

Q33. **[2M]** *ECM 2122 · 2024 Makeup* Explain the operation of accumulator-based computer organization. *(processor organization asked 5×)*

Q34. **[2M]** *ECM 2122 · Nov 2025* Explain the operation of a general register-based computer organization. *(processor organization asked 5×)*

Q35. **[2M]** *ECM 2122 · 2024 Regular* Explain the operation of stack-based computer organization. *(processor organization asked 5×)*

Q36. **[2M]** *ECE 2152 · Jan/Feb 2023* With the help of a block diagram, explain the working of a general register-based processor. — *ECE-variant (processor organization asked 5×)*

Q37. **[2M]** *ECE 2152 · Dec 2022* With the help of a block diagram, explain the working of a stack-based processor. — *ECE-variant*

Q38. **[3M]** *ECE 2152 · Nov/Dec 2023* With the help of a block diagram, explain general register-based computer organization and give suitable example instructions. — *ECE-variant*

Q39. **[3M]** *IT 2123 · Nov/Dec 2023* What is the role of the ALU (Arithmetic Logic Unit) in a CPU, and how does it perform arithmetic and logical operations?

Module 6 — Arithmetic and Logic Unit

Multiplication — Booth's algorithm · division · floating point representation · IEEE standard floating-point representation · floating point arithmetic

Booth's algorithm

Q1. **[4M]** *CSS 2104 · (undated)* Given x = 0101 and y = 1010, compute the product p = xy *using Booth's algorithm. Show all the steps involved.* (Booth's multiplication asked 14×)*

Q2. **[3M]** *CSS 2104 · Nov 2025* Perform 13 × −6 using Booth's algorithm. Write all the necessary steps. *(Booth's multiplication asked 14×)*

Q3. **[4M]** *CSE 2121 · Nov 2025* Determine the product of the following numbers by employing Booth's multiplication algorithm: M = −9 and Q = 15. *(Booth's multiplication asked 14×)*

Q4. **[3M]** *CSE 2121 · Dec 2024* Determine the product of the following numbers by employing Booth's multiplication algorithm: M = −9 and Q = 15. *(Booth's multiplication asked 14×)*

Q5. **[3M]** *CSE 2121 · Nov/Dec 2024* Perform Booth's multiplication algorithm to multiply −12 and −11. Show the sequence of steps clearly and indicate the result. Use the minimum number of bits required. *(Booth's multiplication asked 14×)*

Q6. **[3M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Perform the Booth's multiplication of the following numbers after representing them in 7-bit 2's complement format: M = 33 and Q = −23. *(Booth's multiplication asked 14×)*

Q7. **[5M]** *IT 2123 · Nov/Dec 2023* Multiply $(-7)_{10}$ with $(3)_{10}$ using Booth's multiplication algorithm. Give the flow chart of the multiplication. *(Booth's multiplication asked 14×)*

Q8. **[5M]** *IT 2123 · Dec 2025* Explain Booth's multiplication algorithm with an example and flow diagram. *(Booth's multiplication asked 14×)*

Q9. **[3M]** *ECM 2122 · Dec 2025* Use Booth's algorithm to multiply +23 (multiplicand) by −10 (multiplier), where each number is represented using 6 bits. *(Booth's multiplication asked 14×)*

Q10. **[5M]** *ECM 2122 · 2024 Makeup* Use the Booth algorithm to multiply +23 (multiplicand) by −10 (multiplier), each represented using 6 bits. *(Booth's multiplication asked 14×)*

Q11. **[3M]** *ECM 2122 · Nov 2025* Illustrate the use of Booth's algorithm to multiply −11 (multiplicand) and +8 (multiplier), each represented using 5 bits. *(Booth's multiplication asked 14×)*

Q12. **[5M]** *ECM 2122 · 2024 Regular* Use the Booth algorithm to multiply −11 (multiplicand) by +8 (multiplier), each represented using 5 bits. *(Booth's multiplication asked 14×)*

Q13. **[5M]** *IT 2256 · Jul 2023* Perform the multiplication of $(15)_{10}$ by $(-12)_{10}$ using Booth's multiplication method. *(Booth's multiplication asked 14×)*

Q14. **[5M]** *IT 2256 · May/Jun 2023* Explain and illustrate Booth's algorithm for multiplying $(-7)_{10}$ and $(-5)_{10}$. Show all the steps. *(Booth's multiplication asked 14×)*

Q15. **[2M]** *IT 2256 · 2024* Given M = 17_{10} and Q = -16_{10}, perform multiplication using Booth's algorithm.

Q16. **[5M]** *ECE 2152 · Jan/Feb 2023* By showing all the steps, multiply 14 and −5 using Booth's algorithm. Verify the result using the Modified Booth's algorithm. — *ECE-variant*

Q17. **[2M]** *CSE 2121 / CSE 2151 · Nov 2025* Depict the Hardware Implementation of Multiplication of unsigned Binary Numbers. *(unsigned multiplication hardware asked 2×)*

Q18. **[2M]** *CSE 2151 · Jan/Feb 2023* Depict the Hardware Implementation of Multiplication of unsigned Binary Numbers. *(unsigned multiplication hardware asked 2×)*

Q19. **[2M]** *ECE 2152 · Dec 2022* List the disadvantages of array multipliers. How are they overcome by Booth's and Modified Booth's algorithms? Explain. — *ECE-variant*

Q20. **[3M]** *ECE 2152 · Dec 2022* Design a fast 8 × 8 multiplier using 4 × 4 non-additive multipliers and Wallace tree adders. — *ECE-variant*

Division

Q21. **[3M]** *CSS 2104 · (undated)* Implement the 4×4 restoring integer division algorithm using appropriate register transfer operations. *(restoring division asked 8×)*

Q22. **[5M]** *ICT 2123 · Nov/Dec 2024* Perform division of $(1101)_2$ by $(110)_2$ using the restoring division algorithm, indicating all the steps. *(restoring division asked 8×)*

Q23. **[5M]** *IT 2123 · Jan 2024* Divide 8/3 using the restoring division approach. *(restoring division asked 8×)*

Q24. **[3M]** *CSE 2151 · Jan/Feb 2023* Considering 1010 as Dividend and 0011 as Divisor, perform the division using the restoring division algorithm. *(restoring division asked 8×)*

Q25. **[3M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Using the Restoring Division algorithm, divide the following: divisor = 4 and dividend = 7. *(restoring division asked 8×)*

Q26. **[2M]** *CSE 2121 · Jan 2026* Explain how the Restoring Division Algorithm is used to divide numbers when the dividend = 7 and the divisor = 4. *(restoring division asked 8×)*

Q27. **[5M]** *IT 2256 · 2024* Divide 11 by 3 using the restoring division algorithm indicating all the steps. *(restoring division asked 8×)*

Q28. **[2M]** *CSE 2121 · Dec 2024* Apply Restoring Two's Complement Division to divide −13 by +3. Convert the final answer to its corresponding decimal representation. Show all the necessary steps. *(restoring division asked 8×)*

Q29. **[2M]** *CSE 2121 · Jan 2026* Show a flowchart to illustrate the step-by-step process of the restoring division method in detail. *(restoring division flowchart asked 2×)*

Q30. **[2M]** *CSE 2121 · Nov/Dec 2024* Design a flowchart to illustrate the step-by-step process of the restoring division method in detail. *(restoring division flowchart asked 2×)*

Q31. **[5M]** *ICT 2123 · Dec 2024* Perform division of $(1100)_2$ by $(101)_2$ using the non-restoring division algorithm, indicating all the steps. *(non-restoring division asked 4×)*

Q32. **[4M]** *CSE 2121 · Jan 2026* Evaluate with the Non-Restoring division algorithm to perform the following division: 14/3. *(non-restoring division asked 4×)*

Q33. **[3M]** *CSE 2121 / CSE 2151 · Jan 2024* Apply the Non-Restoring division algorithm to perform the following division: 14/3. *(non-restoring division asked 4×)*

Q34. **[4M]** *ECE 2152 · Nov/Dec 2023* Explain the method of non-restoring division with the help of a flow chart. Divide 7 by 3 using this method. Show all iterations. — *ECE-variant (non-restoring division asked 4×)*

Q35. **[2M]** *IT 2123 · Dec 2025* Explain restoring and non-restoring division with an example.

Q36. **[2M]** *CSE 2151 · Dec 2022* Make use of the paper-and-pencil method to show the division of 22 by 4 in its binary representation. Convert the final answer to its corresponding decimal representation.

IEEE 754 floating point

Q37. **[3M]** *CSS 2104 · (undated)* Perform the addition of the following two 32-bit IEEE 754 single-precision floating-point numbers and represent the result in IEEE 754 format: A = 0 10000010 10100000000000000000000 and B = 0 10000001 11000000000000000000000. Show all steps including conversion to decimal, exponent alignment, addition, normalization and final binary representation. *(IEEE 754 arithmetic asked 11×)*

Q38. **[3M]** *CSS 2104 · Nov 2025* Perform the following in IEEE 32-bit format, showing the result in IEEE 32-bit format: A = 86.625 and B = 10.125. Find A + B. *(IEEE 754 arithmetic asked 11×)*

Q39. **[4M]** *CSE 2121 · Jan 2026* Perform floating point multiplication of the following numbers and represent the result in IEEE 32-bit format: X = 3.344×10^{1} and Y = 8.877×10^{-2}. *(IEEE 754 arithmetic asked 11×)*

Q40. **[3M]** *CSE 2121 / CSE 2151 · Jan 2024* Perform floating point multiplication of X = 3.344×10^{1} and Y = 8.877×10^{-2} and represent the result in IEEE 32-bit format. *(IEEE 754 arithmetic asked 11×)*

Q41. **[3M]** *CSE 2121 · Dec 2024* Convert the numbers A = $(86.625)_{10}$ and B = $(15.125)_{10}$ into IEEE754 32-bit format and perform the multiplication between them. *(IEEE 754 arithmetic asked 11×)*

Q42. **[3M]** *CSE 2121 · Nov/Dec 2024* Convert the decimal numbers 15.75 and 42.5 into IEEE 754 single-precision format. Perform the addition of these two numbers using their IEEE 754 representations and analyse the process to ensure

accuracy. Write the result in both IEEE 754 single-precision format and decimal format. *(IEEE 754 arithmetic asked 11×)*

Q43. **[4M]** *CSE 2151 · Jan/Feb 2023* Consider two decimal numbers A = +81.375 and B = +15.5. (a) Convert to floating point binary. (b) Normalize the number obtained. (c) Represent the normalized numbers in IEEE 32-bit floating point representation. (d) Perform (A / B) using the rules for division of IEEE 32-bit floating point representation, indicating all steps. (e) Store the result in IEEE 32-bit format. *(IEEE 754 arithmetic asked 11×)*

Q44. **[4M]** *CSE 2151 · Dec 2022* Consider two decimal numbers A = +89.125 and B = +15.5. Perform (a)–(d) as above for A / B. *(IEEE 754 arithmetic asked 11×)*

Q45. **[3M]** *CSE 2151 · Dec 2022* Perform the following arithmetic operation using the rules for arithmetic operations in IEEE single precision format: −116.5 − (+211.75). *(IEEE 754 arithmetic asked 11×)*

Q46. **[3M]** *ECE 2152 · Nov/Dec 2023* Add 0.5 and 0.4375 using 32-bit IEEE floating point representation and write the result in 32-bit IEEE format. — *ECE-variant (IEEE 754 arithmetic asked 11×)*

Q47. **[2M]** *CSE 2121 / CSE 2151 · Jan 2024* Express the following number in IEEE 32-bit floating-point format: 1/16. *(IEEE 754 representation asked 4×)*

Q48. **[2M]** *ECE 2152 · Jan/Feb 2023* Convert the following decimal numbers to 32-bit IEEE floating point representation. Write the result in hexadecimal format: (a) 45.125 (b) 0.0125. — *ECE-variant (IEEE 754 representation asked 4×)*

Q49. **[2M]** *ECE 2152 · Dec 2022* Convert the following decimal numbers to 32-bit IEEE floating point representation. Write the result in hexadecimal format: (a) 45.125 (b) 0.0125. — *ECE-variant (IEEE 754 representation asked 4×)*

Q50. **[2M]** *IT 2123 · Dec 2025* Construct the steps for floating-point addition according to the IEEE 754 standard. *(IEEE 754 representation asked 4×)*

Q51. **[4M]** *CSE 2121 · Nov 2025* Evaluate the effectiveness of Chopping and Rounding methods in minimizing truncation errors in floating point representations, and justify which method provides higher numerical accuracy with suitable examples.

Module 7 — Control Unit

Comparison of hardwired and micro-programmed approaches · hardwired control design · Booth's multiplier design · processing section design of Booth's multiplier · Booth's multiplier controller · sequence controller design · micro-programmed control organization · micro-programmed multiplier control unit for Booth's multiplier

Hardwired vs microprogrammed

Q1. **[3M]** *ICT 2123 · Nov/Dec 2024* Explain the difference between hardwired control units and microprogrammed control units. What are the advantages and disadvantages of each? *(hardwired vs microprogrammed asked 5×)*

Q2. **[2M]** *IT 2123 · Nov 2025* Compare hardwired and microprogrammed control units based on flexibility and speed. *(hardwired vs microprogrammed asked 5×)*

Q3. **[2M]** *CSE 2121 · Nov/Dec 2024* Compare and list the pros and cons of both hardwired and microprogrammed control units. *(hardwired vs microprogrammed asked 5×)*

Q4. **[4M]** *CSE 2121 · Nov 2025* In the context of hardwired control unit design, draw the state diagram of the Booth's Multiplier Controller along with the steps in its controller action. In what ways is the Hardwired Control Design Mechanism disadvantageous compared with the Microprogrammed Control Unit Design Mechanism? *(Booth's controller state diagram asked 3×)*

Q5. **[5M]** *CSE 2151 · Jan/Feb 2023* In the context of hardwired control unit design, draw the state diagram of the Booth's Multiplier Controller along with the steps in its controller action. In what ways is the Hardwired Control Design Mechanism disadvantageous compared to the Microprogrammed Control Unit Design Mechanism? *(Booth's controller state diagram asked 3×)*

Q6. **[5M]** *CSS 2104 · Nov 2025* Write the register transfer description algorithm for 4×4 Booth multiplication. Draw the state diagram and indicate all the necessary control signals. Also draw the logic diagram of the Booth's multiplier controller using sequence controller, decoder and counter. *(Booth's controller state diagram asked 3×)*

Hardwired control design

Q7. **[3M]** *CSS 2104 · (undated)* A control circuit has three inputs X, Y and Z and its state diagram is shown below. Design the hardwired controller using counter, decoder and sequence controller. [figure/table not in text] *(hardwired controller design asked 2×)*

Q8. **[5M]** *CSE 2121 / CSE 2151 · Jan 2024* Consider the given state diagram. (i) Draw the block diagram of the controller. (ii) Draw the logic diagram of the controller indicating appropriate counter, decoder and sequence controller. (iii) Derive the truth table of the sequence controller and provide expressions. [figure/table not in text] *(hardwired controller design asked 2×)*

Q9. **[5M]** *IT 2123 · Nov 2025* Design a sequence controller for Booth's multiplier.

Processing section design

Q10. **[4M]** *CSS 2104 · (undated)* Draw the complete processing section diagram for the following register transfer description, indicating all the control points, inputs and outputs. List out the control points and the operation performed. [figure/table not in text] *(processing section design asked 5×)*

Q11. **[3M]** *CSS 2104 · Nov 2025* Design the processing section for the register transfer description given below by establishing the necessary control points. [figure/table not in text] *(processing section design asked 5×)*

Q12. **[4M]** *CSE 2121 · Jan 2026* Design a Register Transfer Level (RTL) description to compute the factorial of a given number using the input bus (INBUS), registers and an adder within the Processing Unit. Clearly identify the control points along with the corresponding actions performed at each stage. Additionally, draw an efficient state diagram indicating the control points and transitions. *(processing section design asked 5×)*

Q13. **[5M]** *CSE 2121 · Dec 2024* Design a Register Transfer Description to find the factorial of a given number using inbus, registers and adder in the Processing Unit. Identify the control points with the corresponding actions and draw an efficient state diagram with corresponding control points. *(processing section design asked 5×)*

Q14. **[5M]** *CSE 2151 · Dec 2022* Design a Register Transfer Description to find x^y. Validate the Register Transfer Description and identify the hardware elements to be used in the Processing Unit. *(processing section design asked 5×)*

Q15. **[3M]** *CSE 2121 · Nov/Dec 2024* A processing section uses three registers X, Y and Z each 4 bits in size, with Inbus and Outbus each 4 lines. Register X must be loaded with Inbus, Register Y cleared to 0, and counter Register Z loaded with 9. It is required to add X and Y iteratively, storing the sum in Y after each iteration, decrementing Z each time.

Q16. **[5M]** *IT 2256 · Jul 2023* Design the processing section for a 4-bit × 4-bit Booth's multiplier. Show all the sequence of steps.

Q17. **[3M]** *ECE 2152 · Jan/Feb 2023* Build the hardware to implement each of the following register transfers. [figure/table not in text] — *ECE-variant (register transfer hardware asked 3×)*

Q18. **[3M]** *ECE 2152 · Dec 2022* Build the hardware to implement each of the following register transfers. [figure/table not in text] — *ECE-variant (register transfer hardware asked 3×)*

Q19. **[4M]** *ECE 2152 · Nov/Dec 2023* Build hardware to implement each of the following register transfers. Assume A, B, C and X are 4-bit registers: if X is even, then … else … [figure/table not in text] — *ECE-variant (register transfer hardware asked 3×)*

Microprogrammed control

Q20. **[3M]** *CSS 2104 · (undated)* Apply the concept of microprogramming to explain how a control unit generates control signals during instruction execution, using a suitable diagram. *(microprogrammed control asked 8×)*

Q21. **[3M]** *CSS 2104 · Nov 2025* Analyze the purpose of the decoder in Wilke's design of a microprogrammed control unit. *(microprogrammed control asked 8×)*

Q22. **[3M]** *CSE 2151 · Jan/Feb 2023* Design a Microprogrammed control unit using Wilke's design for the Register Transfer Description given below. [figure/table not in text] *(microprogrammed control asked 8×)*

Q23. **[5M]** *CSE 2121 · Nov/Dec 2024* Consider the Register Transfer description given below. Draw a neat microprogrammed control unit. Analyze and explain your design with the help of the contents of the condition-select field and actions taken. Write the binary listing of the microprogram for the microprogrammed control unit you designed. [figure/table not in text] *(microprogrammed control asked 8×)*

Q24. **[3M]** *CSE 2121 · Dec 2024* Create the binary listing of the microprogram for the Register Transfer Description given below. Show all the necessary steps. [figure/table not in text] *(microprogrammed control asked 8×)*

Q25. **[5M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Provide the binary listing of the microprogram for the Register Transfer Description given below. Show all the necessary steps. [figure/table not in text] *(microprogrammed control asked 8×)*

Q26. **[3M]** *CSE 2151 · Dec 2022* Design and draw a microprogrammed control organization for the Register Transfer Description given below with necessary details. Show all steps in detail. [figure/table not in text] *(microprogrammed control asked 8×)*

Q27. **[5M]** *IT 2123 · Jan 2024* Design a microprogrammed control unit for a 4×4 Booth's multiplier. *(microprogrammed control asked 8×)*

Q28. **[5M]** *ECE 2152 · Jan/Feb 2023* Design a microprogrammed control unit for Booth's multiplier by giving the RTL code. Explain the control memory organization of Booth's multiplier with a neat diagram. — *ECE-variant (microprogrammed Booth's unit asked 2×)*

Q29. **[5M]** *ECE 2152 · Dec 2022* Design a microprogrammed control unit for Booth's multiplier by giving the RTL code. Explain the control memory organization of Booth's multiplier with a neat diagram. — *ECE-variant (microprogrammed Booth's unit asked 2×)*

Module 8 — Memory Systems and I/O Organization

Cache memories · mapping functions · replacement strategies · I/O device interface · program-controlled I/O · interrupts

Cache mapping and address fields

Q1. **[3M]** *CSS 2104 · (undated)* A block-set-associative cache consists of a total of 128 blocks, divided into 8-block sets. The main memory contains 8192 blocks, each consisting of 256 words. (i) How many bits are there in the main memory address? (ii) How many bits are there in each of the TAG, SET and WORD fields? *(cache address fields asked 9×)*

Q2. **[4M]** *CSS 2104 · Nov 2025* The parameters of a computer memory system are: main memory size = 8K blocks, cache memory size = 512 blocks, block size = 8 words. Determine the size of the tag field of the main memory address under: (i) fully associative mapping (ii) direct mapping (iii) set associative mapping with 16 blocks/set. *(cache address fields asked 9×)*

Q3. **[4M]** *CSE 2121 · Jan 2026* Evaluate the efficiency and address structure of a 2-way set-associative cache having a total capacity of 64 KB and a block size of 128 bytes, when the main memory size is 256 MB. Find the number of bits required for the TAG, SET and WORD (block offset) fields, and how the chosen cache parameters affect address mapping efficiency, hit rate and hardware complexity. *(cache address fields asked 9×)*

Q4. **[4M]** *CSE 2121 · Jan 2026* Consider a block set-associative cache consisting of 64 blocks divided into 4-block sets. The main memory contains 4096 blocks, where each block has 128 words, each word 16 bits long. (1) Illustrate the organization of the cache with a neat labelled diagram. (2) Determine (i) the total number of bits in the main memory (ii) the number of bits required in each of the TAG, SET and WORD fields. *(cache address fields asked 9×)*

Q5. **[5M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Consider a block set-associative cache consisting of 64 blocks divided into 4-block sets. The main memory contains 4096 blocks, each of 128 words of 16 bits length. Depict the organization of the cache through a neat diagram. (i) How many bits are there in main memory? (ii) How many bits are in each of the TAG, SET and WORD fields? *(cache address fields asked 9×)*

Q6. **[2M]** *CSE 2121 / CSE 2151 · Jan 2024* Consider a 4-way set associative mapped cache of size 32 KB with block size 256 bytes. The size of main memory is 128 MB. Find the size of tag, set and word fields. *(cache address fields asked 9×)*

Q7. **[2M]** *ECE 2152 · Jan/Feb 2023* A certain memory system has a 32MB main memory and a 64 KB cache. Blocks are 16 bytes in size. Show the main memory address format to determine the length of TAG, cache line number and WORD fields for the Direct mapping technique. — *ECE-variant (cache address fields asked 9×)*

Q8. **[2M]** *ECE 2152 · Dec 2022* A certain memory system has a 32MB main memory and a 64 KB cache with 16-byte blocks. Show the main memory address format for the Direct mapping technique. — *ECE-variant (cache address fields asked 9×)*

Q9. **[3M]** *ECE 2152 · Nov/Dec 2023* A computer main memory consists of 1024 blocks of 256 words each. The cache memory has eight sets each having four blocks (lines). Determine and show the tag field size of the main memory address for (i) fully associative mapping (and the other techniques). — *ECE-variant (cache address fields asked 9×)*

Q10. **[3M]** *ECE 2152 · Nov/Dec 2023* A set associative cache consists of 64 slots divided into 4-slot sets. Main memory contains 4K blocks of 128 words each. Show the format of the main memory address. — *ECE-variant*

Q11. **[3M]** *IT 2123 · Jan 2024* Explain Set Associative mapping in cache memory with the help of an example.

Q12. **[5M]** *IT 2256 · May/Jun 2023* List the advantages and disadvantages of direct mapping techniques. A computer system uses 16-bit memory addresses and has a 2K-byte cache organized in a direct-mapped manner with 64 bytes per cache block. Assume the size of each memory word is 1 byte. Calculate the fields.

Q13. **[5M]** *ECM 2122 · 2024 Regular* Explain the operation of cache with different mapping techniques.

Cache tracing and replacement

Q14. **[4M]** *CSE 2121 · Jan 2026* A computer system has byte-addressable memory and a data cache capable of storing eight words, where each cache block contains one word. The cache is initially empty. During execution the processor reads data sequentially from the following decimal addresses: 1024, 1025, 1032, 1037, 1024, 1040, 1032, 1025, 1039, 1025, 1037, 1042, 1044, 1037, 1041, 1043. Show the contents of the cache after each access. *(cache tracing asked 5×)*

Q15. **[4M]** *CSE 2121 · Nov 2025* A computer has a small data cache capable of holding eight words and byte-addressable memory, with each cache block consisting of one word (one byte). When a program is executed the processor reads data sequentially from the given decimal addresses. Assume the cache is initially empty. Show the contents of the cache after each memory access for Direct mapped organization. *(cache tracing asked 5×)*

Q16. **[5M]** *CSE 2121 · Dec 2024* A computer has a small data cache holding eight words with byte-addressable memory, each cache block one word. Show the contents of the cache after each memory access for the given sequence of addresses. *(cache tracing asked 5×)*

Q17. **[5M]** *CSE 2151 · Jan/Feb 2023* A computer has a small data cache capable of holding eight words with byte-addressable memory. Show the contents of the cache after each memory access for the given address sequence. *(cache tracing asked 5×)*

Q18. **[5M]** *CSE 2151 · Dec 2022* A computer has a small data cache capable of holding eight words with byte-addressable memory. Show the contents of the cache after each memory access for the given address sequence. *(cache tracing asked 5×)*

Q19. **[5M]** *CSE 2151 · Jan/Feb 2023* A computer has a data cache capable of holding 16 words, each word one byte. The cache block consists of two words and the cache is initially empty. The processor reads data from the given decimal addresses. Show the cache contents.

Q20. **[4M]** *CSE 2151 · Dec 2022* Consider a fully associative cache with 8 cache blocks (0–7). The memory block requests are: 4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7. Outline the steps to demonstrate the working of LRU with a diagrammatic illustration. If the LRU replacement policy is used, what is the result?

Q21. **[3M]** *IT 2123 · Nov 2025* Evaluate the impact of different cache replacement strategies on system efficiency.

Q22. **[3M]** *ECM 2122 · Nov 2025* What is the role of cache in computer memory organization? If a cache has a 95% hit ratio, an access time of 100 ns on a cache hit and 800 ns on a cache miss, compute the effective access time. *(effective access time asked 3×)*

Q23. **[4M]** *CSE 2121 · Jan 2026* Define hit rate and miss penalty. Evaluate the total access time if the miss rate is 0.12, miss penalty is 0.015 ms and cache access time is 10 microseconds. *(effective access time asked 3×)*

Q24. **[5M]** *IT 2256 · 2024* Calculate the average access time experienced by a processor if the cache hit rate is 0.88, the miss penalty is 0.015 milliseconds and the cache access time is 10 microseconds. *(effective access time asked 3×)*

Q25. **[2M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Consider a computer with access times to the cache and main memory of 4τ and 40τ respectively. When a cache miss occurs, a block of 16 words is transferred from main memory to cache. It takes 40τ to transfer the first word of the block, and the remaining words are transferred at a faster rate. Calculate the relevant timings.

Q26. **[3M]** *CSE 2121 / CSE 2151 · Jan 2024* Explain how write-buffers enhance cache accesses in the case of write-through and write-back protocols.

I/O organization and interrupts

Q27. **[5M]** *CSE 2121 · Nov/Dec 2024* Define an interrupt. What is its advantage over program-controlled I/O? Illustrate with a neat diagram how an interrupt request from an I/O device is handled by the processor. *(interrupts asked 7×)*

Q28. **[4M]** *CSE 2151 · Dec 2022* What is an interrupt? Explain the concept of Vectored Interrupts and Interrupt Nesting in the context of handling multiple devices. *(interrupts asked 7×)*

Q29. **[2M]** *CSE 2121 / CSE 2151 · Jan 2024* How are interrupts from multiple devices handled through vectored interrupts? *(interrupts asked 7×)*

Q30. **[2M]** *IT 2123 · Nov/Dec 2023* With neat diagrams, explain the daisy chain technique for servicing multiple interrupts. *(interrupts asked 7×)*

Q31. **[3M]** *IT 2256 · Jul 2023* Write the advantages and disadvantages of daisy-chain-based multiple interrupt handling techniques over polling. *(interrupts asked 7×)*

Q32. **[5M]** *ECE 2152 · Dec 2022* Distinguish the difference between maskable and non-maskable interrupts. Explain with neat diagrams the software polled approach for connecting multiple interrupts to a single interrupt line. List its advantages and disadvantages. — *ECE-variant (interrupts asked 7×)*

Q33. **[5M]** *ECE 2152 · Jan/Feb 2023* Distinguish between maskable and non-maskable interrupts. Explain with neat diagrams software polled approaches for connecting multiple interrupts. — *ECE-variant (interrupts asked 7×)*

Q34. **[5M]** *CSE 2151 · Dec 2022* Summarize the principle involved in the write-through protocol and write-back protocol. Explain the concept of Transfer of Control through the use of interrupts with a diagram.

Q35. **[2M]** *ICT 2123 · Dec 2024* Compare I/O mapped and memory-mapped I/O in terms of address space usage, speed and complexity. Why might one be chosen over the other in a particular system design? *(memory-mapped vs I/O-mapped asked 4×)*

Q36. **[4M]** *CSE 2121 · Nov 2025* Analyze the differences between Memory-Mapped I/O and I/O-Mapped I/O addressing techniques in terms of their addressing method, instruction usage and impact on system performance. *(memory-mapped vs I/O-mapped asked 4×)*

Q37. **[2M]** *CSE 2121 / CSE 2151 · Jan 2024* Compare and contrast Memory-mapped I/O and I/O mapped I/O addressing techniques. *(memory-mapped vs I/O-mapped asked 4×)*

Q38. **[4M]** *ECE 2152 · Nov/Dec 2023* Explain in detail standard I/O and memory-mapped I/O. Also explain the programmed I/O approach for I/O communication. — *ECE-variant (memory-mapped vs I/O-mapped asked 4×)*

Q39. **[5M]** *ECM 2122 · 2024 Makeup* Explain the primary modes through which data is transferred from peripheral devices to the CPU.

Q40. **[2M]** *ICT 2123 · Nov/Dec 2024* What is the role of the CPU in programmed I/O, and how does it differ from direct memory access (DMA)?

Module 9 — Introduction to Parallel Architecture

Pipelining — basic concept · pipeline organization · vector (SIMD) processing · shared-memory multiprocessors · cache coherence

Pipelining

Q1. **[4M]** *CSS 2104 · Nov 2025* What is a pipeline? Demonstrate 5-stage organization of pipelining with a block diagram. *(pipeline organization asked 8×)*

Q2. **[4M]** *CSE 2121 · Jan 2026* Analyze the five-stage pipelining architecture by interpreting its diagram and examining how each interstage buffer contributes to instruction flow, hazard reduction and overall pipeline efficiency. *(pipeline organization asked 8×)*

Q3. **[3M]** *CSE 2151 · Jan/Feb 2023* Give the diagrammatic depiction of the five-stage pipelining architecture and enlist the significance of the various interstage buffers in the architecture drawn. *(pipeline organization asked 8×)*

Q4. **[2M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Why is pipelining widely used in modern computers? Draw the block diagram showing the organization of a pipeline with five stages. *(pipeline organization asked 8×)*

Q5. **[3M]** *IT 2123 · Dec 2025* Apply the concept of pipeline organization to show the role of pipeline registers in improving system performance. *(pipeline organization asked 8×)*

Q6. **[2M]** *CSE 2121 · Jan 2026* What are the critical stages involved in the instruction processing of a pipelined processor, and how do the stages contribute to execution efficiency? *(pipeline organization asked 8×)*

Q7. **[4M]** *CSE 2121 · Nov 2025* Specify the need for pipelining in modern-day processors and list its advantages. Consider a five-stage pipelined processor (IF, ID, EX, MEM, WB) where each stage takes 1 cycle. An instruction sequence consists of five independent instructions. How many cycles will the processor take? *(pipeline speedup asked 4×)*

Q8. **[5M]** *CSE 2121 · Dec 2024* Specify the need for pipelining and list its advantages. For a five-stage pipelined processor with each stage taking 1 cycle and a sequence of five independent instructions, how many cycles will it take? *(pipeline speedup asked 4×)*

Q9. **[2M]** *ECM 2122 · Dec 2025* What is the difference between non-pipelined and pipelined processors? If the number of stages is 6 and the instructions to be executed are 5, how many cycles are required in both cases? *(pipeline speedup asked 4×)*

Q10. **[3M]** *IT 2123 · Dec 2025* Discuss the necessity of implementing pipelined architecture over non-pipelined architecture. *(pipeline speedup asked 4×)*

Q11. **[2M]** *ECE 2152 · Dec 2022* A four-stage pipeline has stage delays of 40, 60, 90 and 70 ns respectively, with a latch delay of 5 ns. Calculate (i) pipeline cycle time (ii) non-pipeline execution time. — *ECE-variant*

Q12. **[3M]** *ECE 2152 · Nov/Dec 2023* A non-pipelined processor operates at a clock rate of 2.5 GHz with an average CPI of five. The processor is upgraded to a six-stage pipelined processor, but the clock speed is reduced to 2 GHz due to internal pipeline delay. Calculate the speedup. — *ECE-variant*

Q13. **[3M]** *IT 2123 · Nov 2025* Explain how pipelining improves performance and discuss hazards in pipelines. *(pipeline hazards asked 5×)*

Q14. **[2M]** *ECM 2122 · Nov 2025* What are the different hazards in pipelining? Explain them with an example for each. *(pipeline hazards asked 5×)*

Q15. **[3M]** *ECM 2122 · 2024 Makeup* What are the different hazards in pipelining? Explain them with an example. *(pipeline hazards asked 5×)*

Q16. **[3M]** *ECM 2122 · 2024 Regular* Explain the following hazards encountered in a pipelined processor: (1) Data hazard (2) Control hazard (3) Structural hazard. *(pipeline hazards asked 5×)*

Q17. **[2M]** *CSE 2121 · Nov/Dec 2024* Consider the given RISC code snippet. Identify the type of data dependency between the two instructions. How can the operand forwarding technique overcome this issue? *(pipeline hazards asked 5×)*

Q18. **[3M]** *CSE 2151 · Dec 2022* What is an Instruction Pipeline? Interpret the effect of a conditional branch on instruction pipelining with a timing diagram.

Shared-memory multiprocessors and cache coherence

Q19. **[3M]** *CSS 2104 · (undated)* (i) Distinguish between Uniform Memory Access and Non-Uniform Memory Access multiprocessors. (ii) Explain how cache coherence is maintained in Write-Through and Write-Back protocols. *(UMA vs NUMA asked 5×)*

Q20. **[2M]** *CSE 2121 · Nov 2025* Distinguish between UMA and NUMA multiprocessors as far as network latency is concerned. *(UMA vs NUMA asked 5×)*

Q21. **[2M]** *CSE 2151 · Jan/Feb 2023* Distinguish between UMA and NUMA multiprocessors as far as network latency is concerned. *(UMA vs NUMA asked 5×)*

Q22. **[3M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Consider the shared memory multiprocessor model used in an advanced computation system. Compare UMA and Non-UMA methods. Justify how these methods impact the performance of the computation with the necessary diagrams. *(UMA vs NUMA asked 5×)*

Q23. **[2M]** *CSE 2121 · Dec 2024* How do recent architectures handle uniform memory and non-uniform memory access issues? Differentiate the features. *(UMA vs NUMA asked 5×)*

Q24. **[5M]** *CSE 2121 / CSE 2151 · Jan 2024* What is the need for cache coherence in a shared-memory multiprocessor architecture? Explain how the Write-back protocol ensures cache coherence. *(cache coherence asked 4×)*

Q25. **[2M]** *CSE 2121 · Nov 2025* Consider a multiprocessor system using a cache coherence protocol. If Processor P1 writes to a cache line and Processor P2 has the same cache line in its cache in the Shared state, what happens to P2's cache line when P1 performs the write? Explain the scenario. *(cache coherence asked 4×)*

Q26. **[2M]** *CSE 2121 · Dec 2024* Consider a multiprocessor system using a cache coherence protocol. If P1 writes to a cache line and P2 has the same cache line in the Shared state, what happens to P2's cache line? Explain. *(cache coherence asked 4×)*

Q27. **[5M]** *IT 2123 · Dec 2025* Discuss the role of SISD in parallel architectures and give real-time examples.

Q28. **[2M]** *CSE 2121 · Jan 2026* Contrast bandwidth and effective throughput of an interconnection network. Justify the need for arbitration when a bus interconnection network is employed in a shared-memory multiprocessing architecture. *(interconnection networks asked 3×)*

Q29. **[2M]** *CSE 2121 · Nov/Dec 2024* Contrast bandwidth and effective throughput of an interconnection network. Justify the need for arbitration when a bus interconnection network is employed in a shared-memory multiprocessing architecture. *(interconnection networks asked 3×)*

Q30. **[3M]** *CSE 2121 · Nov/Dec 2024* Define the bandwidth of an interconnection network. Briefly describe the features and limitations of bus and ring-based interconnection networks. *(interconnection networks asked 3×)*

Q31. **[5M]** *CSE 2151 · Jan/Feb 2023* The Informatics Business unit of Delta Information Solutions operates on shared memory multiprocessors which demand an interconnection requiring a direct link between any pair of units connected to the network, to connect its primary constituent units. [figure/table not in text]

Q32. **[3M]** *CSE 2121 / CSE 2151 · Jan 2024* Briefly explain how the processor and GPU coordinate execution of instructions in a computing system.

Unmapped — Outside the current CES 2104 syllabus

These appeared in the older papers but cover topics not listed in your nine modules.

Transistor-level circuit design (CMOS / TTL)

Your Module 1 starts at the logic-gate level; transistor-level implementation is not in the syllabus. All of these come from the Digital System Design papers.

Q1. **[3M]** *CSE 2123 · Dec 2025* Show the transistor-level diagram for a two-input CMOS NAND gate. Determine why the NMOS transistors are placed in series in the pull-down network while the PMOS transistors are placed in parallel in the pull-up network. *(CMOS gate design asked 7×)*

Q2. **[3M]** *CSE 2123 · Nov 2025* Design a 2-input OR gate using static CMOS logic. Illustrate the primary sources of power dissipation in the resulting CMOS circuit. *(CMOS gate design asked 7×)*

Q3. **[3M]** *CSE 2123 · Nov/Dec 2024* Define the power dissipation in CMOS logic gates. Design a 2-input OR gate using CMOS logic. *(CMOS gate design asked 7×)*

Q4. **[3M]** *CSE 2123 · Dec 2024* Build the equations for low noise margin and high noise margin in logic circuits. Identify the terms present in the equations. Identify the types of power dissipation in logic circuits. *(CMOS gate design asked 7×)*

Q5. **[3M]** *CSE 2153 · Jan/Feb 2023* Construct a NAND gate using a CMOS circuit and explain its operation. *(CMOS gate design asked 7×)*

Q6. **[3M]** *CSE 2153 · Dec 2022* Realize a NAND gate using CMOS technology. *(CMOS gate design asked 7×)*

Q7. **[2M]** *CSE 2153 / CSE 2123 · Jan 2024* Construct a CMOS NOR gate. Write the truth table and transistor states. *(CMOS gate design asked 7×)*

Q8. **[2M]** *ECM 2122 · Nov 2025* Write a switch-level Verilog code for a CMOS inverter using NMOS and PMOS primitives.

Q9. **[3M]** *ECE 2124 · Nov/Dec 2023* Implement a two-input NAND gate with Transistor Transistor Logic (TTL). — *ECE-variant (TTL asked 3×)*

Q10. **[3M]** *ECE 2124 · Jan 2024* Implement a two-input NAND gate with Transistor Transistor Logic (TTL). — *ECE-variant (TTL asked 3×)*

Q11. **[3M]** *ECE 2124 · Dec 2024* What is TTL logic? Explain the working of a 2-input NAND gate using TTL logic. — *ECE-variant (TTL asked 3×)*

Q12. **[3M]** *ECE 2124 · Nov 2025* Explain with an example: (i) noise margin (ii) fan-in in digital circuits (iii) propagation delay. — *ECE-variant (gate parameters asked 2×)*

Q13. **[3M]** *ECE 2124 · Nov/Dec 2024* Explain the following gate parameters: (a) noise margin (b) propagation delay (c) fan-in. — *ECE-variant (gate parameters asked 2×)*

FPGA, ASIC and programmable logic

These come almost entirely from the ECE Digital System Design papers, which substitute FPGA content for your computer-organization modules.

Q14. **[4M]** *ECE 2124 · Nov/Dec 2023* Implement the given circuit using the ACT-2 Sequential module. — *ECE-variant (ACT module implementation asked 7×)*

Q15. **[4M]** *ECE 2124 · Jan 2024* Implement a 3-bit counter using ACT-3 (S-module) FPGA. The logic expressions of the counter are as given. — *ECE-variant (ACT module implementation asked 7×)*

Q16. **[4M]** *ECE 2124 · (undated)* Design and implement a 3-bit Johnson counter using the ACT-2 logic module. — *ECE-variant (ACT module implementation asked 7×)*

Q17. **[4M]** *ECE 2124 · Dec 2024* Implement a full adder using the ACT-1 logic module. — *ECE-variant (ACT module implementation asked 7×)*

Q18. **[4M]** *ECE 2124 · Nov/Dec 2024* Implement a full subtractor using ACT-2 FPGA. — *ECE-variant (ACT module implementation asked 7×)*

Q19. **[4M]** *ECE 2124 · Nov 2025* (a) Implement an SR latch using the ACTEL ACT-1 module. (b) Explain the architecture of the ACT-2 C logic module. — *ECE-variant (ACT module implementation asked 7×)*

Q20. **[3M]** *ECE 2153 · Jan/Feb 2023 & Dec 2022* Implement F = Σm(1,3,5,6) using the ACT-1 module. — *ECE-variant (ACT module implementation asked 7×)*

Q21. **[3M]** *ECE 2124 · Nov/Dec 2023* Implement the given combinational circuit using Xilinx FPGA. Determine the number of CLBs and LUTs required. Show the contents in the SRAM cell. — *ECE-variant (Xilinx CLB/LUT asked 4×)*

Q22. **[3M]** *ECE 2124 · Jan 2024* Implement a 2-bit magnitude comparator using Xilinx FPGA. — *ECE-variant (Xilinx CLB/LUT asked 4×)*

Q23. **[3M]** *ECE 2124 · (undated)* Realize the logic function f(a,b,c,d) = Σm(0,2,6,7,10,14) using a LUT-based FPGA where each LUT has 8-bit RAM. — *ECE-variant (Xilinx CLB/LUT asked 4×)*

Q24. **[4M]** *ECE 2153 · Jan/Feb 2023 & Dec 2022* Implement a 4-bit SISO register using Xilinx CLB. — *ECE-variant (Xilinx CLB/LUT asked 4×)*

Q25. **[3M]** *ECE 2124 · Dec 2024, Nov/Dec 2024 & Nov 2025* Explain the ASIC design flow with a neat flow diagram. — *ECE-variant (ASIC design flow asked 3×)*

Q26. **[3M]** *ECE 2124 · (undated)* Explain the Y-chart representation with respect to FPGA design. Discuss its three domains with neat diagrams. — *ECE-variant*

Q27. **[2M]** *CSS 2104 · Nov 2025* For the following Boolean expressions, draw the PLA: (i) F = ABC' + ABC + A'BC + AB'C + AB'C' (ii) F1 = A'B + BC' and F2 = AB' + A'C. *(PLA/PAL/ROM asked 5×)*

Q28. **[2M]** *ECM 2122 · 2024 Makeup* Implement the following function using PLA: F = AB + BC + ABC. *(PLA/PAL/ROM asked 5×)*

Q29. **[2M]** *ECM 2122 · 2024 Regular* Implement the following function using PAL: F = AB + BC + ABC + CD. *(PLA/PAL/ROM asked 5×)*

Q30. **[4M]** *ECM 2122 · Nov 2025* Implement the following Boolean functions using PAL with four inputs and a 3-wide AND-OR structure: F1(A,B,C,D) = Σm(7,8,9,10,11,12,13,14,15), F2(A,B,C,D) = Σm(1,2,8,12,13). *(PLA/PAL/ROM asked 5×)*

Q31. **[3M]** *ICT 2154 · Jan/Feb 2023* Realize the given Boolean functions using a suitable ROM. *(PLA/PAL/ROM asked 5×)*

Q32. **[3M]** *ECE 2153 · Jan/Feb 2023 & Dec 2022* Implement a BCD to XS-3 code conversion circuit using PROM. — *ECE-variant*

ASM charts and algorithmic state machines

Q33. **[5M]** *CSE 2123 · Dec 2025* Construct an ASM chart for the pseudo code of a bit counter and explain it. *(ASM charts asked 7×)*

Q34. **[5M]** *CSE 2153 · Dec 2022* Construct an ASM chart for the pseudo code of a bit counter and explain it. *(ASM charts asked 7×)*

Q35. **[5M]** *CSE 2153 / CSE 2123 · Jan 2024* Develop the ASM chart for the bit counting circuit which counts the number of 1's in a register. Explain the same. Justify how it is different from a traditional flowchart. *(ASM charts asked 7×)*

Q36. **[5M]** *CSE 2153 · Jan/Feb 2023* Build the algorithm for a bit counting circuit and draw its ASM chart. Differentiate a traditional flow chart and an ASM chart. Also design the datapath circuit and draw the ASM chart for its control circuitry. *(ASM charts asked 7×)*

Q37. **[5M]** *CSE 2123 · Nov/Dec 2024* Design a Bit-Counting circuit to count the number of bits in a register that have the value 1. Draw and explain the Datapath circuit and control circuit. *(ASM charts asked 7×)*

Q38. **[2M]** *CSE 2123 · Dec 2024* Draw an Algorithmic State Machine (ASM) chart to count the number of ones present in a number input to the system. *(ASM charts asked 7×)*

Q39. **[2M]** *CSE 2153 · Jan/Feb 2023* Explain the principal components of an ASM Chart with its symbols. *(ASM charts asked 7×)*

Q40. **[2M]** *CSE 2153 · Dec 2022* List the components in the ASM chart of a Moore type FSM and explain their significance.

Q41. **[2M]** *CSE 2123 · Nov/Dec 2024* Represent the below FSM in the form of an ASM chart. [figure/table not in text]

Quine–McCluskey / tabulation minimisation

Your Module 1 covers K-map simplification only.

Q42. **[5M]** *ICT 2154 · Jan/Feb 2023* Minimize the following using the Tabulation Method and realize using basic gates. [figure/table not in text] *(tabular minimisation asked 2×)*

Q43. **[5M]** *ICT 2154 · Dec 2022* Minimize the following using the Quine-McCluskey Method and realize using basic gates. [figure/table not in text] *(tabular minimisation asked 2×)*

Bus architectures and RALU organization

Q44. **[3M]** *ICT 2123 · Dec 2024* Illustrate and explain the Three-Bus oriented Register ALU architecture and outline the contrasting features with the Two-Bus oriented Register ALU architecture. *(bus architectures asked 6×)*

Q45. **[3M]** *IT 2123 · Nov/Dec 2023* Explain one bus, two bus and three bus oriented RALU with neat diagrams and an example. *(bus architectures asked 6×)*

Q46. **[2M]** *CSE 2121 · Nov 2025* Draw neat diagrams of one bus and three bus architectures. Analyse the number of clock cycles for R2 ← R0+R1 assuming source and destination are general-purpose registers. *(bus architectures asked 6×)*

Q47. **[3M]** *CSE 2121 / CSE 2151 · Jan 2024* Draw neat diagrams of one bus and three bus architectures. Analyse the number of clock cycles assuming source and destination are general-purpose registers. *(bus architectures asked 6×)*

Q48. **[3M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Draw neat diagrams of Two-Bus and Three-Bus architectures. Explain how many clock cycles are required in each to perform an addition operation assuming source and destination are general-purpose registers. *(bus architectures asked 6×)*

Q49. **[3M]** *ECE 2152 · Jan/Feb 2023* A processor has to execute R0 ← R2 + R1. Calculate the number of clock cycles required for (a) single bus RALU (b) two bus RALU (c) three bus RALU. Justify. — *ECE-variant (bus architectures asked 6×)*

Q50. **[3M]** *ECE 2152 · Nov/Dec 2023* Explain the characteristic features of three bus structure organization. Mention its advantages and disadvantages. — *ECE-variant*

Q51. **[3M]** *ECM 2122 · 2024 Makeup* Explain static priority-based bus arbitration protocol. *(bus arbitration asked 2×)*

Q52. **[2M]** *ECM 2122 · 2024 Regular* Explain the lottery-based bus arbitration protocol. *(bus arbitration asked 2×)*

Memory module construction

Your Module 8 covers cache and I/O; building memory modules from chips is not listed.

Q53. **[4M]** *CSE 2121 · Jan 2026* Evaluate the design requirements and organization strategy for constructing a 2G × 16 memory module using 256K × 8 static memory chips. Justify the number of chips required, the memory organization and interconnection scheme, and the advantages and limitations in terms of scalability, performance and hardware complexity. *(memory module design asked 7×)*

Q54. **[3M]** *CSE 2121 · Nov/Dec 2024* Design a memory chip to show the organization of a 2G × 16 memory module using 256K × 8 static memory chips. *(memory module design asked 7×)*

Q55. **[5M]** *CSE 2121 / CSE 2151 · Jan 2024* Design a 2G×32 memory module using 256×8 static memory chips. Draw a neat diagram indicating address lines and data lines and explain its working. *(memory module design asked 7×)*

Q56. **[5M]** *CSE 2121 · Dec 2024* Consider a memory consisting of 64K words of 8 bits each. Organize this memory module using 16K × 1 static memory chips and calculate (i) the number of chips required (ii) the number of rows and columns (iii) how many bits are required for a decoder to select rows. *(memory module design asked 7×)*

Q57. **[3M]** *IT 2256 · 2024* Consider a memory consisting of 64K words of 8 bits each. Sketch the organization to implement this memory using 16K × 1 static memory chips. *(memory module design asked 7×)*

Q58. **[5M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Design a byte-addressable memory module of size 1K × 8 using bit cells with a neat diagram. Calculate the number of address lines and data lines in the final design. *(memory module design asked 7×)*

Q59. **[3M]** *CSE 2151 · Jan/Feb 2023* Design a large memory consisting of 4G words of 64 bits each. Two types of static memory chips are available of the same physical size: 512M × 8 and 1024M × 16. Identify the most suitable memory chip such that a smaller decoder is used and justify. Draw the structure indicating the number of address lines. *(memory module design asked 7×)*

Q60. **[2M]** *CSE 2151 · Dec 2022* With a diagrammatic illustration, discuss the organization of a 2M×32 memory module using 512K×8 static memory chips. Also explain its working.

Q61. **[3M]** *ECE 2152 · Jan/Feb 2023 & Dec 2022* A computer system needs a 1K × 8 RAM. Assuming 256 × 4 RAM chips are available, show how to build the required RAM. Draw a neat logic diagram of your implementation. — *ECE-variant*

Secondary storage and virtual memory

Q62. **[2M]** *CSE 2121 · Jan 2026* Consider a long sequence of accesses to a disk with an average seek time of 4 ms and an average rotational delay of 2 ms. The average block size is 4K bytes and the transfer rate is 34 Mbytes/sec. Assuming randomly located blocks, estimate the average percentage of the total time occupied by seek operations and rotational delays. *(disk timing asked 5×)*

Q63. **[2M]** *CSE 2121 · Dec 2024* Same disk scenario as above (seek 4 ms, rotational delay 2 ms, block 4K bytes, 34 Mbytes/sec). *(disk timing asked 5×)*

Q64. **[2M]** *CSE 2151 · Dec 2022* Consider a long sequence of accesses to a disk with an average seek time of 7 ms and average rotational delay of 4 ms. The average block size is 8K bytes and the transfer rate is 34 Mbytes/sec. Estimate the portion of time occupied by seek and rotational delays. *(disk timing asked 5×)*

Q65. **[5M]** *CSE 2121 · Nov/Dec 2024* Explain how data is organized and accessed in a disk assembly. Consider a disk with average seek time 10 ms, average rotational delay 6 ms, average block size 16K bytes and transfer rate 24 Mbytes/sec. Estimate the average percentage of total time occupied by seek operations. *(disk timing asked 5×)*

Q66. **[5M]** *CSE 2121 / CSE 2151 · Jan 2024* Explain the disk controller and its functions. Consider a disk with an average seek time of 32 ns and a rotational rate of 360 rpm. Each track has 512 sectors and each sector is 512 bytes. What is the time taken to read four continuous sectors? Also determine the data transfer rate. *(disk timing asked 5×)*

Q67. **[3M]** *CSE 2121 · Dec 2024* A disk unit has 16 recording surfaces, 10,000 cylinders, an average of 500 sectors per track and 1024 bytes per sector. (i) What is the maximum number of bytes that can be stored? (ii) What is the data transfer rate in bytes per second at 5400 rpm? *(disk capacity asked 3×)*

Q68. **[2M]** *CSE 2151 · Jan/Feb 2023* A disk unit has 25 recording surfaces, 16,000 cylinders, an average of 200 sectors per track and 512 bytes per sector. What is the maximum number of bytes that can be stored? What is the data transfer rate at 7200 rpm? *(disk capacity asked 3×)*

Q69. **[3M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Illustrate with a diagram how a disk controller is connected with the system bus. Consider a disk unit with 22 recording surfaces, 12,300 cylinders, an average of 320 sectors per track and 512 bytes per sector. (i) What is the maximum number of bytes that can be stored? *(disk capacity asked 3×)*

Q70. **[4M]** *CSE 2121 · Nov 2025* Demonstrate how virtual memory and address translation operate in a computer system by explaining the step-by-step process of translating a virtual address to a physical address using an example memory mapping scenario. *(virtual memory asked 2×)*

Q71. **[5M]** *CSE 2121 · Nov/Dec 2024* Elaborate the working of virtual memory and address translation in detail. *(virtual memory asked 2×)*

DMA

Your Module 8 lists program-controlled I/O and interrupts, but not direct memory access.

Q72. **[4M]** *CSE 2121 · Nov 2025* Explain the concept of DMA data transfer between various I/O devices and memory. Elaborate on the role of the processor and DMA controller with a suitable diagram. *(DMA asked 6×)*

Q73. **[4M]** *CSE 2121 · Nov 2025* Analyse the process of DMA data transfer between I/O devices and memory, explaining each step. Examine the specific roles played by the processor and the DMA controller and illustrate with an appropriate diagram. *(DMA asked 6×)*

Q74. **[5M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Explain how DMA data transfer happens between various I/O devices and memory. Elaborate on the role of the processor and DMA controller with a suitable diagram. *(DMA asked 6×)*

Q75. **[2M]** *IT 2123 · Jan 2024* Explain with a neat diagram the block transfer DMA technique. *(DMA asked 6×)*

Q76. **[3M]** *IT 2256 · May/Jun 2023* How is the block transfer method more efficient in DMA-based I/O data transfer? Explain. *(DMA asked 6×)*

Q77. **[2M]** *IT 2256 · 2024* Define the terms cycle stealing and burst mode with respect to DMA. *(DMA asked 6×)*

Q78. **[2M]** *CSE 2151 · Dec 2022* Compare and contrast I/O Program Controlled Transfer with DMA Transfer.

Advanced architectures and multithreading

Q79. **[5M]** *ECE 2152 · Jan/Feb 2023* Explain in detail the characteristic features of VLIW architecture. Mention its advantages and disadvantages. — *ECE-variant*

Q80. **[5M]** *ECE 2152 · Dec 2022* What is Superscalar architecture? Explain in detail its characteristic features. — *ECE-variant*

Q81. **[2M]** *CSE 2151 · Jan/Feb 2023* Distinguish between Fine-Grained Multithreading and Coarse-Grained Multithreading. *(multithreading asked 3×)*

Q82. **[2M]** *CSE 2121 / CSE 2151 · Nov/Dec 2023* Indicate how processes and threads are differentiated. Explain Coarse-grained and Fine-grained multi-threading. *(multithreading asked 3×)*

Q83. **[3M]** *CSE 2121 · Dec 2024* A processor supports Fine-Grained Multithreading with a cycle-by-cycle thread switch and has four threads, each with independent instructions. Assume each instruction takes 5 cycles to complete and no thread dependencies exist. If each thread begins executing sequentially from the first cycle, how many cycles will it take to complete one instruction from each thread? *(multithreading asked 3×)*

Q84. **[2M]** *ECE 2152 · Jan/Feb 2023* Assume the CSA add time is half of the CPA time. Calculate the total time required to add in a fully parallel 16-operand summation. Find the number of guard bits required. If a Wallace tree structure is used, what is the expression for the total time? — *ECE-variant (Wallace tree / CSA asked 4×)*

Q85. **[3M]** *ECE 2152 · Nov/Dec 2023* Draw the Wallace tree structure for adding 7 single-bit numbers. Give the advantage of having a Wallace structure in the Execution unit. — *ECE-variant (Wallace tree / CSA asked 4×)*

Q86. **[3M]** *ECE 2152 · Nov/Dec 2023* Draw the Wallace tree structure for 7 single-bit operand CSA. Write the expression for finding the number of CSA levels. — *ECE-variant (Wallace tree / CSA asked 4×)*

Q87. **[2M]** *IT 2256 · May/Jun 2023* Explain the addition of four 4-bit binary numbers using the carry save addition concept. *(Wallace tree / CSA asked 4×)*

Q88. **[2M]** *IT 2256 · Jul 2023* A four-bit carry save adder is faster than a carry lookahead adder. TRUE/FALSE — justify with an example.

8086 microprocessor and assembly programming

These come entirely from the IT 2256 "Computer Organization and Microprocessor Systems" papers, which pair computer organization with a microprocessor course your syllabus does not include.

Q89. **[5M]** *IT 2256 · Jul 2023* Explain the execution unit of the 8086 with the help of a neat diagram. *(8086 architecture asked 2×)*

Q90. **[5M]** *IT 2256 · May/Jun 2023* Provide a neat diagram of 8086 architecture and explain the use of various registers of the 8086 in detail. *(8086 architecture asked 2×)*

Q91. **[5M]** *IT 2256 · May/Jun 2023* Illustrate memory and I/O addressing modes available in the 8086 with an example for each.

Q92. **[5M]** *IT 2256 · 2024* Explain the use of the following instructions: (i) RCR (ii) PUSHF (iii) DAS (iv) IMUL (v) JLS.

Q93. **[2M]** *IT 2256 · 2024* If CS=2000h, DS=3000h, SS=4000h, ES=5000h, BX=0030h, determine the physical address for (i) MOV CX, [BX] (ii) ADD AX, 20[BX].

Q94. **[3M]** *IT 2256 · Jul 2023* Explain the following pins in the 8086 processor in detail: (i) HOLD (ii) HLDA (iii) BHE. *(8086 pins asked 2×)*

Q95. **[3M]** *IT 2256 · May/Jun 2023* Illustrate the role of the following pins of the 8086: (i) TEST (ii) IO/M' (iii) RESET. *(8086 pins asked 2×)*

Q96. **[2M]** *IT 2256 · Jul 2023* Explain the following 8086 assembler directives: (i) ORG (ii) DUP (iii) STACK (iv) NEAR. *(assembler directives asked 2×)*

Q97. **[2M]** *IT 2256 · May/Jun 2023* Explain the following assembler directives: (i) ENDP (ii) OFFSET (iii) PROC (iv) DQ. *(assembler directives asked 2×)*

Q98. **[5M]** *IT 2256 · Jul 2023* Write an assembly language program in the 8086 microprocessor to convert an 8-bit BCD number into a hexadecimal number. *(8086 conversion programs asked 3×)*

Q99. **[5M]** *IT 2256 · May/Jun 2023* Write an assembly language program to convert a 4-digit decimal number in the data segment to an equivalent hexadecimal number and store it in the data segment. *(8086 conversion programs asked 3×)*

Q100. **[3M]** *IT 2256 · 2024* Write an assembly language program to convert a 2-digit hexadecimal number in memory into BCD and store the result in memory. *(8086 conversion programs asked 3×)*

Q101. **[2M]** *IT 2256 · Jul 2023* Write an assembly language program to set the trap flag (the 8th bit in the flag register with LSB at position 0). *(trap flag asked 2×)*

Q102. **[2M]** *IT 2256 · 2024* Write a procedure to set the Trap Flag. *(trap flag asked 2×)*

Q103. **[2M]** *IT 2256 · Jul 2023* Write an assembly language program to accept a 2-digit decimal number and display the reversed number.

Q104. **[2M]** *IT 2256 · May/Jun 2023* Provide an assembly language program to display a 2-digit hexadecimal number available in register DL.

Q105. **[2M]** *IT 2256 · May/Jun 2023* Write an assembly language program to perform the equivalent of the AAM instruction.

Q106. **[3M]** *IT 2256 · Jul 2023* Write an assembly language program to display a string using (i) Macro (ii) Procedure.

Q107. **[3M]** *IT 2256 · May/Jun 2023* Write an assembly language program using a recursive procedure to solve nCr, where n and r are both single-digit numbers.

Q108. **[3M]** *IT 2256 · 2024* Write an assembly language program to (i) clear the screen (ii) set the cursor at the centre of the screen.

Q109. **[2M]** *IT 2256 · 2024* Assume a class of people with the grades 69, 87, 96, 45, 75. Develop an assembly language program to find the highest grade.

Q110. **[3M]** *IT 2256 · Jul 2023* The 8254 IC is connected to an 8 MHz 8086 microprocessor. Write a procedure to generate a 100 kHz square wave at OUT0. *(8254 timer asked 3×)*

Q111. **[3M]** *IT 2256 · May/Jun 2023* Explain the Mode 0, Mode 1 and Mode 2 operations of the 8254 with necessary waveforms. *(8254 timer asked 3×)*

Q112. **[5M]** *IT 2256 · 2024* Sketch the steps of Mode 2 operation of the 8254. *(8254 timer asked 3×)*

Q113. **[3M]** *IT 2256 · 2024* Describe the control word format of the 8255.

Q114. **[3M]** *IT 2256 · May/Jun 2023* Explain with a neat diagram the cascading of 8259 ICs.

Q115. **[3M]** *IT 2256 · Jul 2023* Explain what the 8086 does when it encounters divide-by-zero and INTR interrupts at the same time.

Q116. **[2M]** *ECE 2152 · Jan/Feb 2023* With a neat diagram, discuss the timer/counter operation in 8051 microcontrollers. — *ECE-variant*